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# Uncertainty-Aware Decision Making in Multimodal Large Language Models: A Survey of Sources, Signals, Calibration, and Actions

Abderrahmene Boudiaf[a,*], Irfan Hussain[a], Sajid Javed[a]

[a] *Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates*

**Abstract**

Multimodal large language models (MLLMs) increasingly answer questions whose correctness depends on visual, textual, temporal, acoustic, document, chart, or embodied evidence. Their failures are therefore not only linguistic. A fluent answer may conceal poor input quality, a perceptual error, weak grounding, conflict between modalities, unstable reasoning, distribution shift, or a question that is not answerable from the supplied evidence. This survey organizes the literature on uncertainty-aware MLLMs around a decision-centered framework: uncertainty sources give rise to observable signals, signals must be calibrated or controlled for risk, and calibrated uncertainty should determine the system action. We review work on token and logit uncertainty, semantic disagreement, perturbation instability, grounding and attribution scores, verbalized confidence, verifier and judge scores, conformal prediction, selective answering, abstention, clarification, retrieval, self-checking, and escalation. The central argument is that uncertainty should not be evaluated only as a confidence number; it should be evaluated by whether it improves behavior under insufficient, conflicting, shifted, or high-risk multimodal evidence. We position this survey against text-only uncertainty and abstention surveys, broad MLLM surveys, MLLM hallucination surveys, and safety-oriented reviews. We conclude with open problems in source-aware decomposition, action-aware benchmarks, calibration under shift, black-box uncertainty estimation, broader modality coverage, reproducible reporting, and human-centered uncertainty communication.

*Keywords:* multimodal large language models, uncertainty quantification, confidence calibration, selective answering, abstention, answerability, trustworthy AI

## 1. Introduction

Multimodal large language models extend language generation with perceptual and structured inputs, including images, video, audio, charts, documents, and embodied observations. This expansion has made MLLMs useful for visual question answering, chart and document understanding, video reasoning, medical image interpretation, multimodal dialogue, and interactive agents [1, 2]. It has also changed the reliability problem. A generated answer may be linguistically fluent while the underlying evidence is blurred, occluded, incomplete, conflicting, outside the model's training distribution, or simply insufficient to answer the question.

*Corresponding author
*Email address:* 100058322@ku.ac.ae (Abderrahmene Boudiaf)

# 多模态大型语言模型中的不确定性感知决策：来源、信号、校准与行动综述

阿卜杜拉赫曼·布迪亚夫[a,*]，伊尔凡·侯赛因[a]，萨吉德·贾韦德[a]

[a]阿布扎比哈利法理工大学，阿拉伯联合酋长国

**摘要**

多模态大语言模型（MLLMs）越来越多地回答那些其正确性依赖于视觉、文本、时间、声学、文档、图表或具身证据的问题。因此，它们的失败不仅仅是语言上的。流畅的答案可能掩盖输入质量差、感知错误、基础薄弱、多模态冲突、不稳定推理、分布偏移，或问题无法从提供的证据中解答。本文综述了关于不确定感知MLLMs的文献，并以决策为中心的框架进行组织：不确定性来源引发可观察信号，信号必须经过校准或风险控制，而校准后的不确定性应决定系统行动。我们回顾了关于令牌和逻辑不确定性、语义分歧、扰动不稳定性、基础和归因评分、口头化置信度、验证器和评审分数、保形预测、选择性回答、回避、澄清、检索、自检及逃逸等方面的工作。

关键词：多模态大型语言模型、不确定性量化、置信度校准、选择性回答、回避、可回答性、值得信赖的人工智能

## 1. 引言

多模态大语言模型通过感知和结构化输入（包括图像、视频、音频、图表、文档和具体现实中的观察）扩展了语言生成能力。这种扩展使得多模态大语言模型在视觉问答、图表和文档理解、视频推理、医学图像解读以及多模态应用中变得有用。

对话和交互式代理 [1, 2]。它也改变了可靠性问题。生成的答案在语言上可能是流畅的，但其背后的证据可能是模糊的、被遮蔽的、不完整的、相互矛盾的、超出模型训练分布的，或者根本不足以回答问题。

*通讯作者电子邮箱地址：100058322@ku.ac.ae(Abderrahmene Boudiaf)

This distinction matters because uncertainty in MLLMs is not a single phenomenon. Some uncertainty begins in the input: the image may be low quality, the video may omit the relevant moment, or the audio may be noisy. Some uncertainty arises during perception and grounding: the model may recognize the wrong object, attend to the wrong region, or produce a response that is only weakly supported by the visual evidence. Other uncertainty appears later, when the language component fills missing evidence with a prior, when the reasoning path becomes unstable, or when the deployment domain differs from the calibration setting. In the most direct case, the question is not answerable from the supplied multimodal evidence at all.

Existing text-only work on uncertainty quantification, confidence calibration, semantic uncertainty, selective prediction, and abstention provides important foundations [3, 4, 5, 6]. These foundations remain relevant, but they are not sufficient for multimodal systems. A text-only confidence score does not reveal whether the evidence was visible, whether the visual and textual signals agreed, whether the answer was grounded in the correct region, or whether the model should have asked for a clearer input. For MLLMs, uncertainty must be connected to evidence quality and to the action that follows from that evidence.

The central thesis of this survey is therefore decision-centered: uncertainty is useful when it changes what the system does. A system that reports a confidence value but answers every question in the same manner is not yet uncertainty-aware in the deployment sense. A more reliable MLLM should answer when evidence is sufficient, hedge when confidence is limited, abstain or say "I do not know" when evidence is missing, ask for clarification when the user query is underspecified, request a better input when perception is weak, retrieve external evidence when the task requires knowledge beyond the input, self-check when reasoning is unstable, and escalate when the situation is too risky for autonomous handling.

Figure 1 gives the organizing framework for the paper and serves as the survey roadmap. Inputs and context give rise to uncertainty sources; sources produce observable signals; signals are converted into calibrated confidence or controlled risk; and the resulting estimate supports an action policy. This source–signal–calibration–action chain lets us discuss diverse methods without turning the survey into a list of isolated papers.



这一区别很重要，因为多模态大语言模型（MLLMs）中的不确定性并不是单一现象。一些不确定性源于输入：图像可能质量低，视频可能遗漏相关片段，音频可能有噪音。一些不确定性出现在感知和定位阶段：模型可能识别错误的物体，关注错误的区域，或者输出的回答仅有视觉证据的弱支持。其他不确定性出现在更后期，当语言组件用先验信息填补缺失证据时，当推理路径变得不稳定时，或者部署环境与校准环境不同时。在最直接的情况下，从提供的多模态证据根本无法回答问题。

现有关于不确定性量化、置信校准、语义不确定性、选择性预测和回避的纯文本工作提供了重要基础 [3, 4, 5, 6]。这些基础仍然相关，但对于多模态系统而言，它们是不够的。纯文本的置信分数无法揭示证据是否可见，视觉和文本信号是否一致，答案是否基于正确的区域，或者模型是否应该要求更清晰的输入。对于多模态大语言模型，不确定性必须与证据质量以及由该证据引发的行动相关联。

因此，本次调查的核心论点是以决策为中心的：当不确定性能够改变系统的行为时，它才是有用的。一个报告置信值但对每个问题都用相同方式回答的系统，在实际部署意义上尚未具备不确定性感知能力。一个更可靠的大型多模态语言模型（MLLM）应该在证据充分时给出答案，在置信度有限时有所保留，在证据缺失时弃权或说“我不知道”，在用户查询信息不足时请求澄清，在感知薄弱时要求更好的输入，在任务需要输入之外的知识时检索外部证据，在推理不稳定时进行自我检查，以及在情境过于危险以致难以自主处理时升级处理。

图 1 提供了本文的组织框架，并作为调查的路线图。输入和环境情境产生不确定性来源；来源产生可观察的信号；信号被转化为校准的信心或可控的风险；由此产生的估计支持行动策略。这一来源–信号–校准–行动链让我们能够讨论多种方法，而不会将调查变成孤立论文的列表。

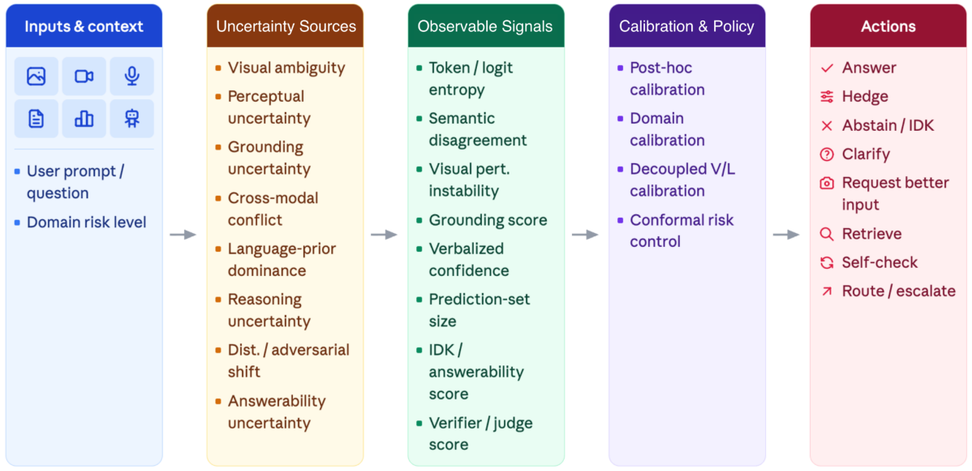



Figure 1: Source–signal–calibration–action framework for uncertainty-aware MLLMs. Multimodal inputs and domain context give rise to uncertainty sources. These sources are observed through model-internal, output-level, evidence-grounding, verbalized, verifier, or judge signals. Calibration and risk-control methods convert raw signals into decision-relevant estimates. The final policy selects an action: answer, hedge, abstain or say IDK, clarify, request better input, retrieve evidence, self-check, or route the case to a stronger model or human expert.

The evidence base around this framework spans calibration before and after multimodal training [7], reasoning-step confidence boundaries [8], decoupled visual and reasoning confidence [9], training-free multimodal uncertainty estimation [10, 11], selective visual question answering [12, 13], multimodal self-awareness and IDK behavior [14, 15], conformal prediction and risk control [16, 17, 18, 19, 20], modality-specific uncertainty benchmarks [21], and refusal or answerability in visual, video, embodied, audio, and omni-modal settings [22, 23, 24, 25, 26, 27, 28, 29, 30]. These studies are related, but they are scattered across calibration, hallucination detection, selective prediction, answerability, conformal risk control, and domain-specific trustworthiness. This survey brings them into one account centered on decision quality.

*Contributions.* The survey makes four contributions. First, it introduces a source–signal–calibration–action taxonomy for uncertainty-aware MLLMs. Second, it synthesizes uncertainty sources that are specific to multimodal systems, including sensory ambiguity, perceptual error, grounding failure, cross-modal conflict, language-prior dominance, reasoning instability, answerability uncertainty, and shift. Third, it connects uncertainty estimation to response actions such as abstention, clarification, retrieval, self-checking, and escalation. Fourth, it reorganizes evaluation around behavior under uncertainty rather than confidence quality alone.

*Scope.* The survey focuses on MLLMs, large vision-language models, and closely related VLM bridge work when the paper directly informs uncertainty estimation, calibration, risk control, answerability, abstention, or action selection. Text-only LLM uncertainty and abstention studies are used as background when they define method families, evaluation concepts, or useful baselines. General hallucination, safety, and broad MLLM surveys are discussed only when they clarify the boundary of the present survey.



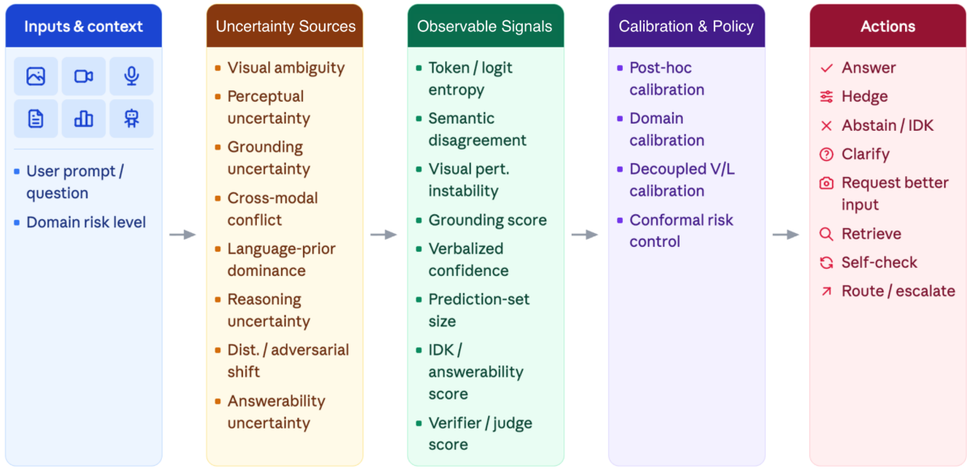



图1：用于不确定感知多模态大语言模型（MLLMs）的源–信号–校准–行动框架。多模态输入和领域上下文产生不确定性来源。这些来源通过模型内部、输出级别、证据基础、口头化、验证器或评判信号进行观测。校准和风险控制方法将原始信号转换为与决策相关的估计值。最终策略选择一个行动：回答、保留意见、放弃或说不知道（IDK）、澄清、请求更好的输入、检索证据、自检，或将案例转交给更强的模型或人工专家。

围绕该框架的证据基础涵盖了多模态训练前后的校准 [7], 、推理步骤置信界限 [8], 、视觉与推理置信的解耦 [9], 、无需训练的多模态不确定性估计 [10, 11], 、选择性视觉问答 [12, 13],、多模态自我意识与“我不知道”行为 [14, 15], 、保序预测与风险控制 [16, 17, 18, 19, 20], 、特定模态的不确定性基准 [21],„ 以及在视觉、视频、具身、音频和全模态环境中的拒答或可答性[22, 23, 24, 25, 26, 27, 28, 29, 30]。这些研究彼此相关，但分散在校准、幻觉检测、选择性预测、可答性、保序风险控制以及特定领域的可信性等方面。本综述将它们汇聚在以决策质量为中心的统一叙述中。

贡献。该综述有四个贡献。首先，它为不确定感知多模态大语言模型（MLLM）引入了一个源–信号–校准–行动分类法。其次，它综合了特定于多模态系统的不确定性来源，包括感官模糊、感知错误、落地失败、跨模态冲突、语言先验支配、推理不稳定、可回答性不确定性和偏移。第三，它将不确定性估计与响应行动联系起来，如放弃、澄清、检索、自检和升级。第四，它重新组织了评估，关注的是不确定性下的行为，而不仅仅是置信度质量。

范围。该调查关注多模态大语言模型（MLLMs）、大型视觉-语言模型（VLMs）以及直接涉及不确定性估计、校准、风险控制、可回答性、回避或行动选择的相关视觉-语言模型桥接工作。当文本-only的大语言模型（LLM）不确定性和回避研究定义了方法类别、评估概念或有用的基线时，这些研究将作为背景使用。一般性的幻觉、安全性以及广泛的MLLM调查仅在它们有助于澄清本调查边界时进行讨论。

*Organization.* Section 2 defines the scope, positioning, and evidence base. Section 3 reviews sources of uncertainty in MLLMs. Section 4 organizes observable signals and estimation methods. Section 5 discusses calibration and risk control. Section 6 connects uncertainty to response actions. Section 7 reviews evaluation protocols, benchmarks, and metrics. Section 8 discusses deployment settings. Section 9 gives the research agenda. Sections 10 and 11 close the paper.

## 2. Scope, Positioning, and Evidence Base

A survey on uncertainty-aware MLLMs needs clear boundaries. We define uncertainty-aware decision making as the study of methods that estimate, express, calibrate, or control uncertainty in order to improve behavior under multimodal evidence. The definition includes uncertainty estimation, but it also includes the downstream policy: whether the system should answer, hedge, abstain, ask for clarification, request better input, retrieve evidence, verify its response, or escalate the case.

### *2.1. Problem definition and terminology*

Let a user provide multimodal evidence $M$, a prompt or task $q$, and optional context $c$ such as domain, risk level, or retrieved information. An uncertainty-aware MLLM does not only produce an answer $y$. It also estimates a signal $s$, converts that signal into a calibrated confidence or risk estimate $r$, and selects an action $a$ through a policy $\pi$. Informally, the decision process is

$$(a, y, r) = \pi\big(q, M, c, s(q, M, c)\big). \tag{1}$$

The answer may be returned directly, qualified with a hedge, replaced by an IDK response, delayed until clarification or retrieval is performed, or routed to another system or human expert. This formulation is deliberately simple. Its purpose is to keep the survey focused on the relation between uncertainty and action rather than on a single mathematical definition of uncertainty.

We use *uncertainty source* for the underlying reason a response may be unreliable. We use *uncertainty signal* for an observable quantity or behavior that may reveal unreliability, such as entropy, logit margin, semantic disagreement, visual perturbation instability, grounding score, verbalized confidence, prediction-set size, answerability score, or verifier score. We use *calibration* for the alignment between an estimated confidence or risk and empirical correctness or action cost. We use *policy* for the rule that maps uncertainty to a response action.

### *2.2. Inclusion criteria*

A paper is central to this survey when uncertainty is part of the method, benchmark, metric, or response policy. This includes MLLM and LVLM work on confidence estimation, calibration, selective answering, conformal prediction, hallucination or error detection when framed as uncertainty, answerability, IDK behavior, refusal, self-checking, retrieval decisions, and escalation. We also include bridge work from VLMs and text-only LLMs when it defines concepts or methods that are necessary for understanding MLLM uncertainty.

We do not attempt to survey every paper on multimodal hallucination, safety, robustness, or broad MLLM evaluation. Those literatures are adjacent and often relevant, but they become central here only when they connect evidence uncertainty to calibrated confidence, risk control, or action selection. This boundary prevents the survey from becoming a general MLLM reliability review.



组织。第2节定义了范围、定位和证据基础。第3节回顾了MLLM的不确定性来源。第4节组织了可观察信号和估计方法。第5节讨论了校准和风险控制。第6节将不确定性与响应行动联系起来。第7节回顾了评估协议、基准和指标。第8节讨论了部署环境。第9节给出了研究议程。第10节和第11节结束本文。

## 2. 范围、定位与证据基础

关于不确定性感知多模态大语言模型（MLLMs）的调查需要明确的界限。我们将不确定性感知决策定义为研究旨在估计、表达、校准或控制不确定性的方法，以在多模态证据下改善行为。该定义包括不确定性估计，但也包括下游策略：系统是否应回答、保留意见、回避、请求澄清、请求更好的输入、检索证据、验证其响应或升级处理案例。

### *2.1.* 问题定义和术语

让用户提供多模态证据 $M$、提示或任务 $q$，以及可选的上下文 $c$，例如领域、风险水平或检索的信息。不确定性感知的多模态大语言模型不仅生成一个答案 $y$。它还会估计一个信号 $s$，将该信号转换为校准后的置信度或风险估计 $r$，并通过策略 $\pi$ 选择一个动作 $a$。非正式地说，决策过程是

$$(a, y, r) = \pi\big(q, M, c, s(q, M, c)\big). \tag{1}$$

答案可以直接返回，带有保留语，也可以被 IDK 回答替代，延迟直到进行澄清或检索，或者转发给其他系统或人工专家。这种表述刻意简单。其目的是保持调查集中在不确定性与行动之间的关系，而不是集中于不确定性的单一数学定义上。

我们使用不确定性来源来表示响应可能不可靠的根本原因。我们使用不确定性信号来表示可能揭示不可靠性的可观察量或行为，例如熵值、对数边际、语义分歧、视觉扰动不稳定性、基准分数、口头化信心、预测集大小、可回答性评分或验证器评分。我们使用校准来表示估计的置信度或风险与实际正确性或行动成本之间的一致性。我们使用策略来表示将不确定性映射到响应动作的规则。

### *2.2.* 纳入标准

当不确定性是方法、基准、度量或响应策略的一部分时，论文在本次调查中具有核心地位。这包括关于置信度估计、校准、选择性回答、符合性预测、幻觉或错误检测的多模态大语言模型（MLLM）和大视觉语言模型（LVLM）工作，当这些工作以不确定性、可回答性、IDK 行为、拒绝、自我检查、检索决策和升级的形式呈现时。我们还包括从视觉语言模型（VLMs）和仅文本大语言模型衍生的桥梁工作，当它定义的概念或方法对于理解 MLLM 不确定性是必要时也会被纳入。

我们并不打算对关于多模态幻觉、安全性、鲁棒性或广泛 MLLM 评估的每篇论文进行全面综述。这些文献是相关且经常有参考价值的，但只有当它们将证据不确定性与校准信心、风险控制或行动选择联系起来时，它们才在此处成为核心内容。这一界限防止本综述变成一般性的 MLLM 可靠性回顾。

### *2.3. Position relative to adjacent surveys*

The closest survey families cover text-only uncertainty, text-only abstention, MLLM hallucination, MLLM safety, or broad MLLM evaluation. These areas are important, but they do not fully center the source-to-action chain for multimodal evidence. Table 1 summarizes the positioning in categorical terms. This format is preferable to an arbitrary two-axis plot because the difference is conceptual rather than geometric.

Table 1: Positioning against adjacent survey families. The table separates related coverage from the gap addressed by this survey. Citations in the first four rows are representative anchors for adjacent survey families; the final row states the positioning of the present survey.

| **Survey family** | **Main coverage** | **Gap for this survey** |
|---|---|---|
| Text-only LLM uncertainty | Uncertainty quantification, confidence calibration, semantic uncertainty, and selective prediction in language-only settings [3, 4, 5, 31]. | Limited treatment of visual evidence quality, grounding, cross-modal conflict, and multimodal answerability. |
| Text-only abstention and IDK | Knowledge boundaries, refusal, and selective answering in language-only settings [6]. | Abstention is not tied to visual sufficiency, image quality, or multimodal grounding. |
| MLLM hallucination and safety | Hallucination symptoms, attack and defense settings, benchmark construction, and safety risks [32, 33, 34, 35]. | Often organized around failure detection or safety risk rather than uncertainty signals, calibration, and action policies. |
| Broad MLLM and evaluation surveys | Architectures, training data, capabilities, benchmarks, and application coverage [1, 2]. | Too broad to compare uncertainty sources, signal access assumptions, calibration targets, and response actions in detail. |
| **This survey** | Source-aware uncertainty, observable signals, calibration, risk control, selective answering, abstention, clarification, retrieval, self-checking, escalation, and action-aware evaluation. | Integrates multimodal uncertainty estimation with decision policy. |

### *2.4. Evidence base and traceability*

The core evidence base contains 67 focused evidence rows drawn from MLLM and LVLM studies, VLM bridge work, domain-specific studies, and directly relevant surveys. The manuscript also uses additional background survey citations for positioning in Table 1; these background citations are not counted as focused evidence rows. The evidence matrix retains one additional non-counted bridge row for traceability. Search logs, evidence-tier accounting, extraction fields, and corpus-refresh details are retained in Appendix A. This separation preserves traceability without interrupting the main synthesis. In the main synthesis tables, citations attached to row labels are representative anchors rather than exhaustive lists; explanatory, action, and caveat cells synthesize recurring patterns from the cited works and the surrounding subsections.

With the scope fixed, the rest of the paper follows the framework in Figure 1. We first identify where uncertainty enters the multimodal pipeline, then ask how those sources can be observed, calibrated, converted into actions, and evaluated.



### *2.3.* 与相邻测量的相对位置

最接近的调查类别涵盖了纯文本不确定性、纯文本回避、多模态大语言模型（MLLM）幻觉、MLLM 安全性或广泛的 MLLM 评估。这些领域很重要，但它们并没有完全以多模态证据的源到动作链为中心。表 1 用分类术语总结了这些定位。这种格式比任意的双轴图更可取，因为差异是概念性的，而不是几何性的。

表1：相对于相邻调查系列的定位。该表将相关覆盖与本调查所解决的空白分开。前四行的引用是相邻调查系列的代表性参考点；最后一行说明了本调查的定位。

| **调查家庭** | **主要 coverage** | **此调查的空白** |
|---|---|---|
| 纯文本大型语言模型不确定性 | 不确定性量化，置信度校准，语义不确定性，以及选择性仅语言预测设置 [3, 4, 5, 31]。 | 视觉的有限处理，证据质量，依据，跨模态冲突，以及多模态可回答性 |
| 仅文本弃权我不知道 | 知识边界，拒绝，以及选择性回答仅语言设置 [6]。 | 弃权与视觉无关充足性、图像质量，或多模态基础 |
| 多模态大语言模型幻觉和安全 | 幻觉症状，发作以及防御设置，基准施工和安全风险 [32, 33, 34, 35]。 | 经常围绕失败组织而是检测或安全风险比不确定信号，校准和行动政策。 |
| 广泛的多语言大型模型和评估调查 | 架构，训练数据，能力、基准以及应用覆盖 [1, 2]。 | 太宽泛而无法比较不确定性来源，信号访问假设，校准目标和响应详细行动。 |
| **此调查** | 源感知不确定性，可观测信号，校准，风险控制，有选择的回答，弃权，澄清，检索，自我检查、升级，以及动作感知评估 | 整合多模态不确定性带有决策策略的估计。 |

### *2.4.* 证据基础与可追溯性

核心证据库包含从MLLM和LVLM研究、VLM桥梁工作、特定领域研究以及直接相关的调查中提取的67行重点证据。手稿还使用了额外的背景调查引用，以便在表1中进行定位；这些背景引用不计入重点证据行。证据矩阵保留了一行额外的非计数桥梁行以保持可追溯性。搜索日志、证据层级记录、提取字段和语料库更新细节保存在附录A中。这种分离在不干扰主要综合分析的情况下保持了可追溯性。在主要综合分析表中，附加到行标签的引用是代表性锚点，而非详尽列表；解释、操作和注意事项单元格综合了被引用作品及其相关子部分中的重复模式。

在范围确定后，论文的其余部分遵循图1中的框架。我们首先确定不确定性在多模态流程中进入的位置，然后探讨这些来源如何被观察、校准、转化为行动以及评估。

## 3. Sources of Uncertainty in MLLMs

The first question in the framework is where uncertainty comes from. For MLLMs, this question cannot be answered by looking only at the final answer or token distribution. The uncertainty may originate in the input, the perceptual encoder, the cross-modal alignment, the reasoning path, the deployment domain, or the answerability of the task. A source-aware taxonomy matters because different sources call for different actions. Blur may call for a better image. Missing evidence may call for abstention. Cross-modal conflict may call for self-checking or retrieval. A high-risk shifted domain may call for escalation.

Figure 2 provides a visual intuition for this idea. It is schematic rather than dataset-specific. Its role is to show that the same user question can require different response policies: clear evidence supports direct answering, degraded evidence supports hedging or a request for better input, occluded evidence supports stating uncertainty or abstaining, and referent ambiguity supports clarification or disambiguation.

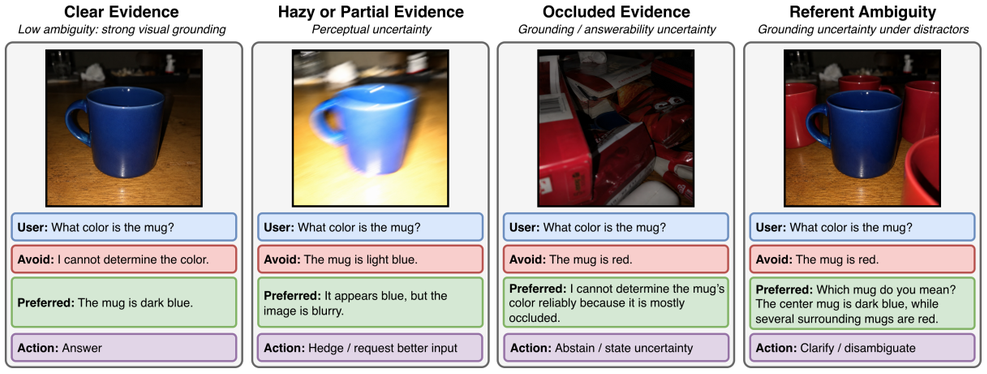



Figure 2: **From evidence conditions to uncertainty-aware actions.** The same user question can require different model behavior depending on the quality, availability, and grounding of the visual evidence. With clear evidence, direct answering is appropriate, and unnecessary abstention becomes a failure mode. With hazy or partially degraded evidence, the model should hedge or request better input rather than over-specifying an uncertain visual attribute. With occluded evidence, a confident answer may reflect weak grounding or distractor bias, so the safer behavior is to state uncertainty or abstain. With referent ambiguity, the visual input contains multiple plausible targets, so the model should clarify or disambiguate the referred object instead of answering from a salient distractor. The examples are illustrative rather than exhaustive.

### *3.1. Sensory and perceptual uncertainty*

Sensory and perceptual uncertainty occurs before high-level reasoning. The input may be blurred, low-resolution, cropped, occluded, noisy, temporally sparse, or visually ambiguous. The model may also misrecognize an object, miss a small detail, fail to read text in an image, confuse a chart axis, or lose information across video frames. In these cases, additional language reasoning may amplify the error rather than repair it. The action should therefore depend on whether the evidence can be improved. A model can request a clearer image, ask which object the user means, use a specialized OCR or perception module, or abstain when the relevant evidence is absent.



## 3. MLLMs中的不确定性来源

框架中的第一个问题是，不确定性来自何处。对于多模态大语言模型（MLLMs）来说，仅通过观察最终答案或词元分布无法回答这个问题。不确定性可能来源于输入、感知编码器、跨模态对齐、推理路径、部署领域或任务的可回答性。一个源感知的分类法很重要，因为不同的来源需要不同的处理措施。模糊可能需要更好的图像。缺失证据可能需要弃权。跨模态冲突可能需要自我检查或信息检索。高风险的迁移领域可能需要升级处理。

图2为这一想法提供了直观的理解。它是示意性的，而不是特定数据集的。其作用是表明相同的用户问题可能需要不同的响应策略：明确的证据支持直接回答，证据不充分支持谨慎回答或请求更好的输入，证据被遮挡支持表明不确定或选择回避，而指代模糊支持进行澄清或消歧。

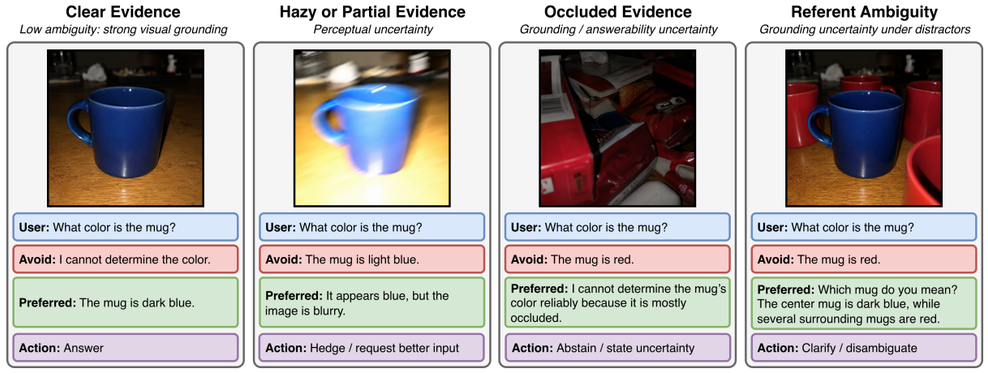



图2：从证据条件到不确定性感知的行为。同一个用户问题可能根据视觉证据的质量、可获得性和依据情况而需要不同的模型行为。当证据清晰时，直接回答是合适的，不必要的回避会成为一种失败模式。当证据模糊或部分退化时，模型应当采取保守态度或请求更好的输入，而不是对不确定的视觉属性进行过度具体化。当证据被遮挡时，自信的回答可能反映出基础薄弱或干扰偏差，因此更安全的做法是表明不确定性或回避。当指称存在歧义时，视觉输入包含多个可能的目标，因此模型应当澄清或消除指称对象的歧义，而不是从显眼干扰项中作答。这些示例是说明性而非详尽的。

### *3.1.* 感觉与知觉的不确定性

感觉和感知的不确定性发生在高层次推理之前。输入可能是模糊的、低分辨率的、裁剪过的、被遮挡的、有噪声的、时间上稀疏的，或视觉上模糊的。模型也可能错误识别一个物体、漏掉一个小细节、无法读取图像中的文字、混淆图表轴，或在视频帧之间丢失信息。在这些情况下，额外的语言推理可能会放大错误而不是修复它。因此，行动应取决于证据是否可以改进。模型可以请求更清晰的图像、询问用户指的是哪个物体、使用专门的 OCR 或感知模块，或者在相关证据缺失时选择放弃。

### 3.2. Grounding and cross-modal uncertainty

Grounding uncertainty asks whether the answer is supported by the supplied multimodal evidence. Cross-modal uncertainty asks whether the modalities agree. A prompt may mention an object that is not visible, an image may contradict the user's wording, retrieved text may conflict with visual evidence, or the model may produce a plausible answer that is not anchored to the relevant region. Hallucination studies often capture the failure symptom, while uncertainty-aware methods ask whether the system can detect the weak support before committing to an answer [32, 33, 36]. The natural actions are self-checking, evidence citation, retrieval, abstention, or escalation depending on the risk of the task.

### 3.3. Language-prior and reasoning uncertainty

MLLMs inherit strong priors from their language components. These priors are helpful when evidence is complete, but risky when evidence is ambiguous or missing. The model may infer a common object that is not visible, complete a scene using stereotypical co-occurrence, or produce a fluent explanation for a detail that the input does not support. Reasoning uncertainty appears when intermediate steps are unstable, when sampled chains of thought lead to different conclusions, or when a correct perception is followed by an incorrect inference. Methods that examine reasoning boundaries, self-consistency, verifier scores, or semantic disagreement can expose this class of uncertainty [8, 10, 11].

### 3.4. Answerability uncertainty

Answerability uncertainty is the case where the user asks for information that is not available from the multimodal input. The image may not show the requested object, the video may not contain the relevant event, the chart may not encode the requested quantity, or the prompt may require external facts that were not supplied. Treating this as ordinary error is not enough. A model should learn when the right answer is no answer, or when the correct next step is to ask for missing evidence. Work on unsolvable problem detection, multimodal self-awareness, visual IDK behavior, and visually unanswerable questions treats answerability as a target in its own right [22, 14, 15, 24].

### 3.5. Distributional, adversarial, and deployment uncertainty

Calibration is conditional on the data and context where it was measured. A model calibrated on common web images may not remain calibrated on medical scans, scientific plots, screenshots, low-light video, robotics observations, or culturally specific imagery. Distribution shift changes what a confidence value means. Adversarial or misleading inputs create a deliberate version of the same issue by making the model confident in the wrong evidence. Under these conditions, the policy may need stricter thresholds, conformal risk control, retrieval, routing to a specialized model, or human escalation [7, 9, 16, 17].

### 3.6. Interactions among sources

The sources above are separable for analysis, but they are not independent in practice. Poor input quality can cause perceptual error. Perceptual error can produce weak grounding. Weak grounding can be hidden by a strong language prior. Cross-modal conflict can destabilize reasoning. Distribution shift can break calibration. Missing evidence can make both confidence and explanation misleading. Figure 3 summarizes



### 3.2. 基础与跨模态不确定性

基础不确定性是指答案是否得到了提供的多模态证据的支持。跨模态不确定性是指各模态之间是否一致。提示可能提到一个不可见的物体，图像可能与用户的措辞相矛盾，检索到的文本可能与视觉证据冲突，或者模型可能生成一个看似合理但没有锚定到相关区域的答案。幻觉研究通常捕捉的是失败的症状，而关注不确定性的方法则在给出答案前询问系统是否能够检测到证据的薄弱支持 [32, 33, 36]。自然的操作包括自我检查、引用证据、检索、弃权或升级，具体取决于任务的风险。

### 3.3. 语言优先与推理不确定性

多模态大语言模型（MLLMs）继承了其语言组件中的强先验。这些先验在证据完整时非常有用，但在证据模糊或缺失时则存在风险。模型可能推断出不可见的常见物体，使用刻板的同时出现模式来补全场景，或者为输入不支持的细节生成流畅的解释。当中间步骤不稳定、采样的思维链导致不同结论，或者正确的感知之后出现错误的推理时，就会出现推理不确定性。检查推理边界、自洽性、验证器评分或语义分歧的方法可以揭示这一类不确定性 [8, 10, 11]。

### 3.4. 可答复性不确定性

可回答性不确定性是指用户请求的信息无法从多模态输入中获得的情况。图像可能未显示所请求的对象，视频可能不包含相关事件，图表可能未编码所请求的数量，或者提示可能需要未提供的外部事实。仅将其视为普通错误是不够的。模型应该学会何时正确的答案是不回答，或者何时正确的下一步是请求缺失的证据。对不可解决问题检测、多模态自我意识、视觉“我不知道”（IDK）行为以及视觉不可回答问题的研究，将可回答性本身作为一个目标 [22, 14, 15, 24]。

### 3.5. 分布式、不确定性对抗和部署不确定性

校准取决于所测数据和背景。在常见网页图像上校准的模型，可能在医学扫描、科学图表、屏幕截图、弱光视频、机器人观测或特定文化的图像上不再保持校准。分布变化会改变置信值的含义。对抗性或误导性输入通过让模型对错误证据保持自信，从而制造了同样问题的故意版本。在这些情况下，策略可能需要更严格的阈值、符合性风险控制、检索、路由到专门模型或人工升级。

### 3.6. 来源之间的相互作用

上述来源在分析上是可分的，但实际上它们并不独立。输入质量差会导致感知错误。感知错误可能产生薄弱的基础。薄弱的基础可能被强大的语言先验掩盖。跨模态冲突可能使推理不稳定。分布偏移可能破坏校准。缺失证据可能使信心和解释都具有误导性。图3总结了这些内容。

these recurring interactions, and Table 2 gives the source vocabulary used in the rest of the survey.

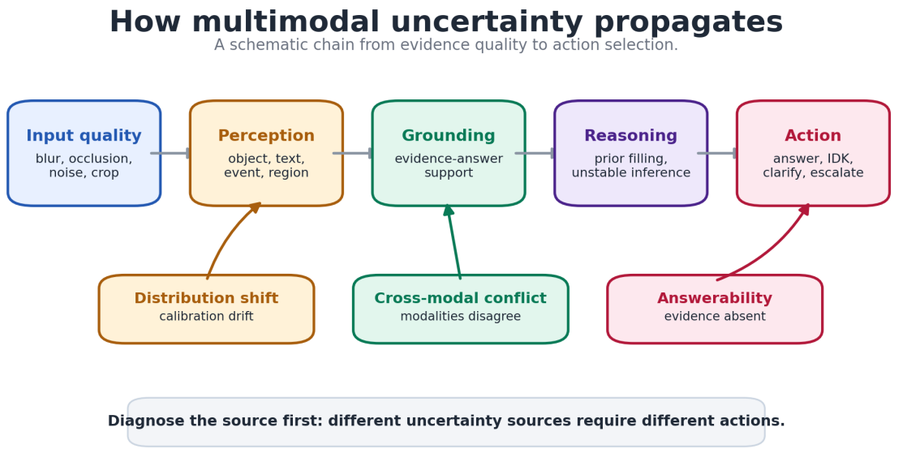



Figure 3: **Propagation and interaction of MLLM uncertainty sources.** The figure illustrates how uncertainty can cascade across sensory, grounding, reasoning, and policy layers in multimodal language models. In the sensory–grounding chain, ambiguous visual input can lead to perceptual uncertainty, weaken the alignment between evidence and claims, and allow language priors to fill visual gaps. These failures can become entangled with cross-modal conflict, where text and image evidence disagree, and with reasoning uncertainty, where inference becomes unstable under weak or conflicting evidence. System-level uncertainty can also arise from distributional or adversarial conditions that cause calibration failure, while missing or absent evidence creates answerability uncertainty and should trigger abstention or clarification rather than forced answering. Black arrows denote primary causal pathways, gray arrows denote entanglement between uncertainty sources, and colors distinguish perceptual-chain components, reasoning components, systemic sources, and model actions. The diagram is a schematic synthesis rather than an exhaustive causal model.



这些反复出现的互动，表 2 给出了在调查其余部分中使用的词汇来源。

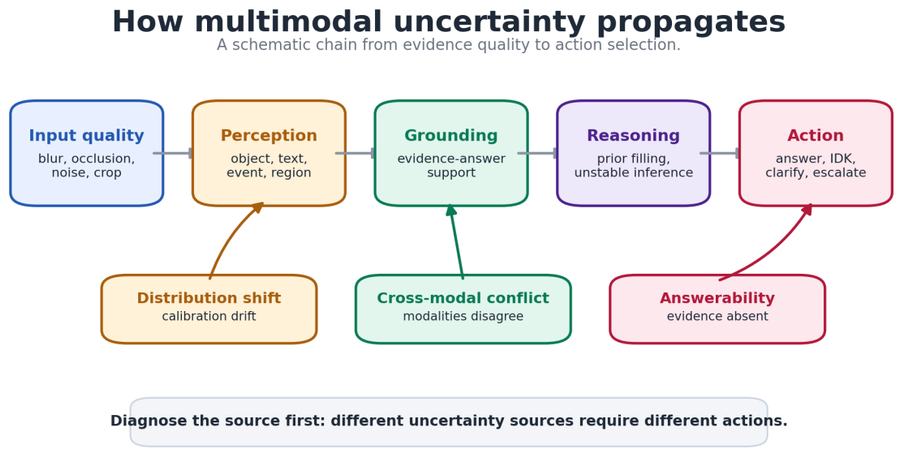



图3：MLLM不确定性来源的传播与相互作用。该图说明了不确定性如何在多模态语言模型的感知、落地、推理和策略层之间层层传递。在感知-落地链中，模糊的视觉输入可能导致感知不确定性，削弱证据与陈述之间的对齐，并允许语言先验填补视觉缺口。这些失败可能与跨模态冲突交织在一起，即文本和图像证据不一致；也可能与推理不确定性交织在一起，即在证据薄弱或矛盾时推理变得不稳定。系统级不确定性也可能源于分布性或对抗性条件导致的校准失败，而缺失或不存在的证据会产生可回答性不确定性，应触发放弃或澄清，而不是强制回答。黑色箭头表示主要因果路径，灰色箭头表示不确定性来源之间的交织，颜色区分感知-c

Table 2: Multimodal uncertainty sources and decision implications. This table defines the source taxonomy used in later sections. Citations in the source column are representative anchors; the signal and action columns are author synthesis from the cited evidence and surrounding discussion.

| Source | What goes wrong | Possible signal | Likely action |
|---|---|---|---|
| Sensory and perceptual uncertainty [14, 37, 38, 39] | The relevant evidence is degraded, ambiguous, occluded, noisy, too small, temporally sparse, or misrecognized. | Visual perturbation instability, low perceptual confidence, inconsistent captions, OCR or layout failure. | Request better input, clarify the target, route to a specialized perception module, or abstain. |
| Grounding and cross-modal uncertainty [40, 36, 27, 41] | The answer is weakly supported by the evidence, or the text, image, retrieved context, or other modalities disagree. | Grounding score, attribution mismatch, hallucination detector, semantic disagreement, verifier or judge score. | Self-check, cite evidence, retrieve, abstain, or escalate. |
| Language-prior and reasoning uncertainty [8, 10, 11, 42] | The model fills visual gaps with prior knowledge, or an unstable reasoning path changes the conclusion. | Sampling disagreement, reasoning-step confidence, semantic variation, verifier score, answer variation. | Self-check, sample alternatives, use tools, retrieve, or escalate. |
| Answerability uncertainty [22, 14, 15, 24] | The question cannot be answered from the supplied input or requires missing evidence. | Answerability classifier, IDK score, prediction-set size, refusal consistency, unsupported-evidence flag. | Abstain, say IDK, ask for missing evidence, or clarify. |
| Distributional, adversarial, and deployment uncertainty [7, 9, 16, 17] | The model faces domain shift, modality shift, corrupted input, misleading prompts, adversarial cues, or high-risk deployment conditions. | Calibration drift, conformal risk estimate, perturbation disagreement, shift detector, safety or domain-risk score. | Apply risk control, abstain, retrieve, route, or escalate. |

This source vocabulary gives the rest of the survey a stable reference point. The next question is not whether uncertainty exists, but how a system can observe it before committing to a response.

## 4. Observable Signals and Estimation Methods

The sources in Section 3 are mostly latent. A blurred image, a weak grounding path, a misleading prompt, or a missing piece of evidence does not automatically announce itself to the user. An uncertainty method is therefore a way of making one of these latent conditions observable. We use *signal* broadly: a signal may come from token probabilities, hidden states, repeated samples, perturbation behavior, grounding checks, verbalized confidence, verifier scores, judge models, or conformal score functions.



表2：多模态不确定性来源及决策影响。本表定义了后续章节中使用的来源分类法。来源栏中的引用是代表性参考点；信号栏和行动栏是作者根据所引用的证据及相关讨论的综合整理。

| 源 | 出了什么问题 | 可能的信号 | 可能的行动 |
|---|---|---|---|
| 感官和感知的不确定性 [14, 37, 38, 39] | 相关证据被降级，模糊的，遮挡的，嘈杂，太小，时间上稀疏，或者识别错误。 | 视觉扰动不稳定，低感知信心，不一致的字幕，光学字符识别或布局失败。 | 请求更好的输入，澄清目标，通往专科的路线感知模块，或弃权。 |
| 接地与跨模态不确定性 [40, 36, 27, 41] | 答案是软弱地由……支持证据，或者文本，图像，已检索上下文，或其他方式不一致。 | 接地评分归因不匹配，幻觉检测器，语义分歧，验证者或评判分数。 | 自我检查，引用证据，取回，弃权，或升级。 |
| 语言优先和推理不确定性 [8, 10, 11, 42] | 模型填充视觉与先前的差距知识，或者一个不稳定的推理路径更改结论。 | 采样分歧，推理步骤置信度，语义变体，验证器得分，回答变体。 | 自检，样本替代方案，使用工具，检索，或升级。 |
| 可回答性不确定性 [22, 14, 15, 24] | 这个问题不能从……得到回答提供的输入或需要缺失证据。 | 可回答性分类器，IDK 分数，预测集大小，拒绝一致性不支持的证据旗. | 弃权，说不知道，提问因缺少证据，或澄清。 |
| 分布式的对抗性的，并且部署不确定性 [7, 9, 16, 17] | 模型面临领域转移，模态移位，损坏的输入，误导性提示，对抗性线索，或高风险部署条件。 | 校准漂移保形风险估计，扰动分歧，转变探测器、安全或域名风险评分。 | 应用风险控制弃权，取回，路由，或升级。 |

这个源词汇为调查的其余部分提供了一个稳定的参考点。下一个问题不是不确定性是否存在，而是一个系统在做出回应之前如何观察它。

## 4. 可观察信号与估计方法

第3节中的来源大多是潜在的。一张模糊的图片、一条薄弱的依托路径、一条误导性的提示，或缺失的证据，并不会自动向用户显现。因此，不确定性方法是一种使这些潜在条件变得可观察的方式。我们广义地使用信号：信号可能来自标记概率、隐藏状态、重复样本、扰动行为、依托检查、口头表达的置信度、验证器评分、评判模型或符合性评分函数。

A useful signal should answer three questions. First, what access does it require? Some methods need logits, hidden states, visual tokens, or gradients, while others operate on black-box outputs. Second, what source of uncertainty does it approximate? Entropy, for example, may indicate label uncertainty, but it may also mix perception, reasoning, and answerability. Third, what action can it support? A signal that detects poor visual evidence should support different behavior from a signal that detects unstable reasoning. Figure 4 summarizes this access–signal–diagnosis–action view.

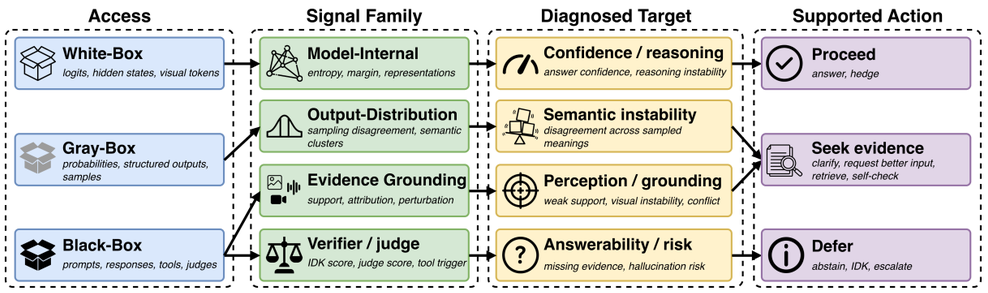



Figure 4: **Choosing uncertainty signals under model-access constraints.** The figure maps model access to observable uncertainty signals, diagnosed targets, and supported actions. White-box access enables model-internal signals; gray-box access supports output-distribution signals; and black-box access relies on prompts, responses, tools, and judge-based signals. These signals diagnose confidence, semantic instability, grounding failure, or answerability risk, which then guide whether the system should proceed, seek additional evidence, or defer. Arrows indicate typical rather than exclusive pathways.

### *4.1. Model-internal signals*

Model-internal signals use information exposed by the model before or during decoding. The most familiar examples are token probabilities, logit margins, entropy, sequence likelihood, and maximum softmax probability. In vision-language settings, these signals can be extended to hidden representations, visual tokens, attention patterns, or modality-specific features. They are attractive because they are direct and often cheap once model access is available.

Several lines of work illustrate the value of internal signals for response control. First-token logit distributions have been used to detect cases where a large vision-language model should withhold a response, including unanswerable or unsafe cases [43]. Reasoning-step confidence has been used to improve knowledge-boundary awareness in MLLMs [8]. Hidden representations have also been probed for abstention in OCR-heavy visual question answering, where the model may generate plausible but incorrect text readings [38].

Internal signals become more multimodal when they diagnose the contribution of the visual input itself. Dropout Decoding estimates uncertainty in visual tokens and masks uncertain tokens during generation [37]. Vision-aware self-evaluation methods estimate how much the image contributes to the answer, for example through image information scores or core-region masking [44]. Blind-image contrastive ranking compares real-image and blacked-out-image behavior so that confidence reflects visual grounding rather than language priors alone [45]. These methods move beyond generic confidence and ask whether the answer is actually using the visual evidence.

The limitation is access. Many deployed MLLMs expose only text, not logits or hidden states. Internal scores can also be hard to interpret: high entropy does not identify



一个有用的信号应该回答三个问题。首先，它需要什么样的访问权限？一些方法需要 logits、隐藏状态、视觉令牌或梯度，而其他方法则在黑箱输出上运行。第二，它近似哪种不确定性来源？例如，熵可能表示标签的不确定性，但它也可能混合感知、推理和可回答性。第三，它可以支持什么样的行动？一个检测到较差视觉证据的信号应该支持不同的行为，而不是一个检测到不稳定推理的信号。图4总结了这一访问–信号–诊断–行动视角。

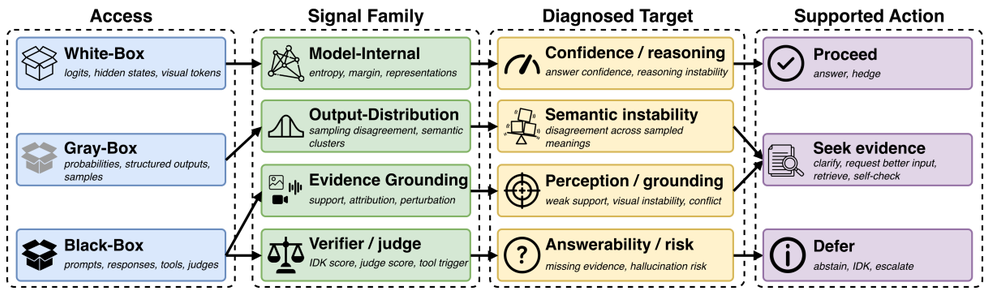



图4：在模型访问受限下选择不确定性信号。该图将模型访问映射到可观察的不确定性信号、诊断目标和支持的操作。白盒访问可以使用模型内部信号；灰盒访问支持输出分布信号；黑盒访问依赖提示、响应、工具和基于评判的信号。这些信号用于诊断置信度、语义不稳定性、基础失败或可回答性风险，从而指导系统是否应继续、寻求额外证据或延迟操作。箭头表示典型路径，而非排他路径。

### *4.1.* 模型内部信号

模型内部信号使用模型在解码前或解码过程中暴露的信息。最常见的例子包括标记概率、逻辑回归边际、熵、序列可能性以及最大 softmax 概率。在视觉-语言环境中，这些信号可以扩展到隐藏表示、视觉标记、注意模式或特定模态的特征。一旦可以访问模型，它们很有吸引力，因为它们直接且通常成本低廉。

几项工作展示了内部信号在响应控制中的价值。首先，已经使用单词标记的对数几率分布来检测大型视觉-语言模型在何种情况下应当拒绝响应，包括不可回答或不安全的情况 [43]。推理步骤的置信度已被用于提高多模态大语言模型（MLLMs）在知识边界上的意识 [8]。还通过对隐藏表示进行探测，用于在以光学字符识别（OCR）为主的视觉问答中进行弃权判定，在这种情况下，模型可能会生成看似合理但不正确的文本识别结果[38]。

当内部信号在诊断视觉输入本身的贡献时，它们变得更加多模态。Dropout 解码在视觉标记中估计不确定性，并在生成过程中屏蔽不确定的标记 [37]。视觉感知的自我评估方法估计图像对答案的贡献程度，例如通过图像信息分数或核心区域屏蔽 [44]。盲图像对比排序比较真实图像和遮挡图像的行为，使得置信度反映视觉基础，而不仅仅是语言先验 [45]。这些方法超越了通用置信度的概念，探讨答案是否实际上使用了视觉证据。

限制在于访问。许多部署的 MLLM 仅公开文本，而不公开 logits 或隐藏状态。内部分数也可能难以解释：高熵无法识别

whether uncertainty comes from perception, grounding, reasoning, or answerability. For this reason, internal signals are powerful but rarely sufficient as the only basis for action.

### *4.2. Sampling and semantic-disagreement signals*

Sampling-based signals estimate uncertainty by asking whether the model remains stable under repeated generation, paraphrased prompts, equivalent inputs, or nearby visual-question variants. If a model gives semantically incompatible answers to equivalent cases, the disagreement itself becomes an uncertainty signal. This family is especially useful when logits and hidden states are unavailable.

Selective and consistency-oriented VQA methods ask whether visually grounded responses remain reliable enough to accept, while related robustness analyses examine how visual reasoning changes under equivalent or perturbed inputs [12, 13, 46]. VL-Uncertainty estimates hallucination risk from response variation under semantically equivalent visual and textual perturbations, clustering similar responses and measuring uncertainty over the clusters [47]. Uncertainty-o extends this idea into a model-agnostic multimodal perturbation framework for exposing epistemic uncertainty [48]. FESTA uses functionally equivalent and complementary sampling to assess trust in MLLM responses across visual and audio reasoning settings [11]. Multi-model approaches, such as semantic opinion pooling, estimate system-level uncertainty by comparing semantically aligned outputs across several VLMs [49].

The strength of this family is practical access: repeated outputs are often available even when the model is closed. The weakness is cost and ambiguity. Sampling can be expensive, and semantic diversity is not always error. A model may produce several correct paraphrases, or it may repeatedly produce the same wrong answer because the same language prior dominates each sample. Sampling signals are therefore more reliable when combined with grounding, answerability, or calibration checks.

### *4.3. Evidence-grounding and perturbation signals*

MLLM uncertainty is often evidence uncertainty. A response can be fluent, confident, and wrong because it is not supported by the image, chart, video, audio, or document. Evidence-grounding signals ask whether the answer can be traced to the supplied input. They include region-grounding scores, attribution scores, visual engagement measures, core-region masking, image perturbation sensitivity, OCR or chart-consistency checks, cross-view consensus, and contradiction detection across modalities.

Grounding has been used directly to calibrate uncertainty estimates by combining self-consistency with visual grounding confidence [40]. Clinical and radiology settings use attribution or repeated-report consistency to flag unsupported statements for abstention or extra review [50, 51]. OCR-specific work uses latent probes or cross-view verifiability to decide when generated text should be accepted or rejected [38, 20]. Other work studies evidence collapse, where the model gives low-entropy but visually ungrounded answers, motivating signals that combine language confidence with visual engagement [42].

Grounding signals also matter when modalities conflict. Benchmarks and analyses of cross-modal contradiction, omni-modal dissonance, and modality preference show that a model may follow the wrong modality or hide conflict behind a fluent answer [27, 41, 52]. In these cases, uncertainty is not simply low confidence; it is evidence disagreement. The appropriate action may be to clarify, retrieve, self-check, or abstain rather than to average the modalities into one answer.



无论不确定性来自感知、基础、推理还是可回答性。因此，内部信号虽然强大，但作为行动的唯一依据却很少足够。

### *4.2.* 采样与语义分歧信号

基于采样的信号通过询问模型在重复生成、改写提示、等效输入或邻近的视觉-问题变体下是否保持稳定来估计不确定性。如果模型对等效情况给出了语义上不兼容的答案，那么这种分歧本身就成为不确定性信号。这一类方法在日志值和隐藏状态不可用时尤其有用。

选择性和一致性导向的VQA方法会询问视觉基础的回答是否仍然足够可靠以被接受，而相关的鲁棒性分析则检查在等效或扰动输入下视觉推理如何变化 [12, 13, 46]。VL-Uncertainty通过在语义等效的视觉和文本扰动下对回答变化进行评估，估计幻觉风险，对相似回答进行聚类并测量聚类的不确定性 [47]。Uncertainty-o将这一思想扩展到一个模型无关的多模态扰动框架中，以揭示认知不确定性 [48]。FESTA使用功能等效和互补采样，在视觉和音频推理场景中评估对MLLM回答的信任度 [11]。多模型方法，例如语义意见汇聚，通过比较多个VLM上语义对齐的输出，估计系统层面不确定性 [49]。

这个家族的优势在于实际可用性：即使模型关闭了，也经常可以获得重复的输出。弱点是成本和模糊性。采样可能很昂贵，并且语义多样性并不总是错误。一个模型可能会生成几个正确的改写，或者因为相同的语言先验主导每次采样而反复生成相同的错误答案。因此，当采样信号与基础事实、可回答性或校准检查结合使用时，更加可靠。

### *4.3.* 证据基础与扰动信号

MLLM的不确定性往往是证据的不确定性。一个回答可能流畅、自信却错误，因为它没有得到图像、图表、视频、音频或文档的支持。证据归因信号用于判断答案是否可以追溯到提供的输入。它们包括区域归因评分、归因评分、视觉参与度测量、核心区域遮罩、图像扰动敏感性、OCR或图表一致性检查、跨视图共识，以及跨模态矛盾检测。

通过将自洽性与视觉定位置信度结合，接地已被直接用于校准不确定性估计 [40]。临床和放射学环境使用归因或重复报告一致性来标记不支持的陈述以便弃权或额外审核 [50, 51]。OCR特定的工作使用潜在探针或跨视图可验证性来决定生成的文本何时应被接受或拒绝 [38, 20]。其他工作研究了证据塌缩，即模型给出低熵但视觉上未定位的答案，这促使使用结合语言置信度与视觉参与的信号 [42]。

当模态发生冲突时，落地信号也很重要。对跨模态矛盾、全模态不协调和模态偏好的基准测试和分析表明，模型可能会遵循错误的模态，或者在流畅的回答背后隐藏冲突 [27, 41, 52]。在这些情况下，不确定性不仅仅是低信心；它是证据不一致。适当的做法可能是澄清、检索、自我检查或弃答，而不是将各模态平均成一个答案。

### *4.4. Verbalized confidence, IDK scores, verifiers, and judges*

When internal access is limited, the model can be asked to express its confidence, identify what it does not know, or evaluate its own answer. Verbalized confidence is attractive because it is easy to collect and naturally communicates with users. It is also risky because the model's stated confidence may not match empirical correctness. Studies of verbalized calibration in VLMs and of token-versus-verbal confidence mismatch show that implicit confidence and explicit self-assessment can diverge [53, 54].

IDK and answerability signals are a more action-oriented form of self-assessment. Instead of asking only "how confident are you?", these methods ask whether the model should answer at all. Benchmarks for perceptual self-awareness, visual IDK behavior, unanswerable visual questions, and uncertainty-aware self-assessment make the model's knowledge boundary observable [14, 15, 24, 55]. These signals are useful because they connect directly to abstention, refusal, or clarification.

Separate verifiers, judge models, ensembles, and agentic controllers provide another route. A verifier may check grounding, compare candidate answers, call external tools, or route the case to another model. These signals can support richer policies than a single confidence score. They also introduce another reliability problem: the verifier or judge may itself be miscalibrated. Work on VLM judges, for example, distinguishes the ability to rank outputs from the ability to provide calibrated scalar scores [56]. Judge-based uncertainty should therefore be treated as an estimate to calibrate, not as ground truth.

### *4.5. Choosing a signal under access constraints*

No signal family dominates all settings. The right choice depends on model access, modality, uncertainty source, and action cost. White-box systems can exploit logits, hidden states, and visual-token features. Gray-box systems can use structured outputs, repeated generations, or tool traces. Black-box systems must rely on sampling, verbalized confidence, external verifiers, judge models, and interaction with the user. Table 3 summarizes the main design trade-offs.



### *4.4.* 口头表达的信心、*IDK* 分数、验证者和评审

当内部访问受限时，可以要求模型表达其自信度、识别其不知道的内容或评估自己的答案。口头化的自信度具有吸引力，因为它易于收集，并且能够自然地与用户沟通。但它也存在风险，因为模型所声明的自信度可能与实际正确性不符。对视觉语言模型（VLM）中口头化校准的研究以及对标记与口头自信度不匹配的研究表明，隐性自信和显性自我评估可能会出现偏离 [53, 54]。

IDK 和可回答性信号是一种更注重行动的自我评估形式。这些方法不仅仅问“你有多自信？”，而是询问模型是否应该回答。用于感知自我意识、视觉 IDK 行为、无法回答的视觉问题以及不确定性自我评估的基准，使模型的知识边界可被观察到 [14, 15, 24, 55]。这些信号很有用，因为它们直接与回避、拒绝或澄清相关。

独立的验证器、评审模型、集成模型和代理控制器提供了另一种途径。验证器可以检查依据基础信息的准确性、比较候选答案、调用外部工具，或将案例转给另一个模型。这些信号可以支持比单一置信度得分更丰富的策略。它们也引入了另一个可靠性问题：验证器或评审模型本身可能存在校准不准的情况。例如，对视觉语言模型（VLM）评审模型的研究，将输出排序能力与提供校准标量分数的能力区分开来。因此，基于评审的的不确定性应被视为一种需要校准的估计，而不是事实真相。

### *4.5.* 在访问限制下选择信号

没有哪种信号家族能在所有环境中占主导地位。正确的选择取决于模型访问权限、模态、不确定性来源以及行动成本。白箱系统可以利用logits、隐藏状态和视觉令牌特征。灰箱系统可以使用结构化输出、重复生成或工具痕迹。黑箱系统必须依赖采样、口头化的置信度、外部验证器、评判模型以及与用户的互动。表3总结了主要的设计权衡。

Table 3: Signal and method taxonomy for uncertainty-aware MLLMs. The table is organized by what information is available, what uncertainty source the signal mainly diagnoses, and what action it can support. Citations in the signal-family column provide representative support for each method family.

| Signal family | Typical access | Primary diagnosis | Action value | Main caveat |
|---|---|---|---|---|
| Token, logit, and entropy signals [43, 8, 54] | Logits or probabilities | Decoding uncertainty, hidden knowledge boundary, label ambiguity | Confidence reporting, selective answering, refusal triggers | Often unavailable in closed models and weak at separating uncertainty sources. |
| Hidden-state and visual-token signals [37, 44, 45, 38] | Open weights, embeddings, attention, or visual-token features | Perception, visual grounding, OCR or region-level uncertainty | Request better input, route to perception module, abstain from ungrounded answers | Requires model access and may be difficult to compare across architectures. |
| Sampling and semantic disagreement [47, 48, 11, 49, 13] | Repeated outputs, perturbed prompts, or equivalent inputs | Response instability, reasoning uncertainty, epistemic uncertainty | Self-checking, selective prediction, abstention, verifier selection | Expensive and sensitive to sampling settings; repeated agreement can still be confidently wrong. |
| Grounding, attribution, and perturbation checks [40, 50, 51, 42, 20] | Region evidence, attribution, grounding model, or input perturbation | Evidence support, hallucination risk, cross-modal conflict | Evidence citation, retrieval, better-input requests, abstention, escalation | Depends on the quality of the grounding or perturbation procedure. |
| Verbalized confidence and IDK scores [53, 54, 14, 15, 55] | Prompt access and generated text | Self-reported confidence, answerability, knowledge boundary | IDK, clarification, user-facing uncertainty communication | Verbal confidence may be miscalibrated or strategically phrased. |
| Verifier, judge, ensemble, and agentic signals [56, 57, 58, 59, 60] | Extra model, judge, tool, or controller | Cross-model disagreement, route quality, answer support | Retrieval, self-checking, routing, escalation | The judge or controller also requires calibration and can introduce bias. |
| Conformal and risk-control scores [16, 17, 18, 19, 20] | Calibration set and nonconformity score | Empirical coverage, prediction-set size, action risk | Risk-controlled answering, abstention, set prediction | Risk statements depend on calibration data, exchangeability, and the chosen score. |

The transition from signals to actions is not automatic. A signal may rank risky outputs well but still give confidence values that are too high or too low. It may also be calibrated for one target, such as final-answer accuracy, while being used for another target, such as grounding or refusal. Calibration and risk control address this gap.

## 5. Calibration and Risk Control

Calibration asks whether a reported confidence or score has the meaning that the policy assumes. Risk control asks whether the resulting policy can keep error, violation, or unsupported-answer rates within an acceptable range. Both are needed because raw uncertainty signals are not decision-ready. A score that is useful for ranking errors may still be poorly calibrated; a verbal confidence statement may sound careful while being empirically unreliable; and a grounding score may detect unsupported claims without specifying the risk of answering.

### *5.1. Calibration targets*

The first design choice is the target. MLLM calibration can target final-answer correctness, step-level reasoning correctness, visual evidence support, grounding faithfulness, answerability, prediction-set coverage, hallucination risk, or downstream action cost. These targets should not be treated as interchangeable. A model can be calibrated on multiple-choice VQA accuracy and still be overconfident when the image is unanswerable.



表3：面向不确定性感知多模态大语言模型（MLLMs）的信号和方法分类。表格按照可获得的信息、信号主要诊断的不确定性来源以及其可以支持的操作进行组织。信号类别列中的引用为每个方法类别提供了代表性的支持。

| 信号系列 | 典型访问 | 主要诊断 | 动作值 | 主要警告 |
|---|---|---|---|---|
| 标记、对数几率，以及熵信号 [43, 8, 54] | 逻辑回归或概率 | 解码不确定性，隐秘的知识边界，标签歧义 | 置信度报告选择性回答，拒绝触发 | 经常不可用于封闭模型和不擅长分离不确定性来源。 |
| 隐藏状态和视觉标记信号 [37, 44, 45, 38] | 自由重量嵌入，注意，或者视觉标记特征 | 感知，视觉接地、光学字符识别或区域级不确定性 | 请求更好的输入，通向感知的路线模块，避免没有根据的答案 | 需要模型访问并可能是难以比较跨架构。 |
| 采样和语义分歧 [47, 48, 11, 49, 13] | 重复输出，不安的提示，或者等效输入 | 响应不稳定，推理不确定性，认知不确定性 | 自我检查，选择性预测，弃权，验证者选择 | 昂贵且对抽样敏感设置；重复协议仍然可以自信而错误。 |
| 接地归因，以及扰动检查 [40, 50, 51, 42, 20] | 区域证据，归因基础模型，或输入扰动 | 证据支持幻觉风险，跨模态冲突 | 证据引用检索，更好的输入请求，弃权，升级 | 取决于……的质量接地或扰动程序。 |
| 口头表达的自信和 IDK 分数 [53, 54, 14, 15, 55] | 提示访问并生成 text | 自我报告的信心可回答性，知识边界 | 我不知道，需要澄清面向用户的不确定性沟通 | 口头自信可能校准不当或有策略地措辞。 |
| 核实者，裁判，合奏，以及能动的信号 [56, 57, 58, 59, 60] | 额外模型，法官，工具，或者控制器 | 跨模型分歧，路线质量，答案支持 | 检索，自我检查，路由，升级 | 法官或控制器也需要校准并且可以介绍偏见。 |
| 共形和风险控制评分 [16, 17, 18, 19, 20] | 校准集and不墨守成规得分 | 经验覆盖率预测集大小，行动风险 | 风险可控回答，弃权，设置预测 | 风险说明取决于校准数据，可交换性，以及所选分数。 |

从信号到行为的转变并不是自动的。一个信号可能对风险输出的排序很准确，但仍然给出过高或过低的置信值。它可能也只针对一个目标进行校准，例如最终答案的准确率，而被用于另一个目标，例如依据性或拒绝。校准和风险控制可以解决这一差距。

## 5. 校准与风险控制

校准是询问报告的置信度或分数是否具有策略所假定的意义。风险控制是询问由此产生的策略是否能够将错误、违规或无支持答案的概率保持在可接受范围内。两者都是必要的，因为原始的不确定性信号尚不能直接用于决策。一份对错误排序有用的分数可能仍然校准不佳；一份口头的置信声明听起来可能很谨慎，但在经验上不可靠；而一个依据分数可能能够检测无支持的主张，却无法说明答题的风险。

### *5.1.* 校准目标

第一个设计选择是目标。MLLM 校准可以针对最终答案的正确性、步骤级推理的正确性、视觉证据支持、基础真实性、可回答性、预测集合覆盖率、幻觉风险或下游操作成本。这些目标不应被视为可以互换的。一个模型可以在多项选择 VQA 准确性上进行校准，但在图像无法回答时仍可能过度自信。

A model can be calibrated for answer correctness and still fail to communicate whether the answer is grounded. Clear target definition is therefore part of the method, not a reporting detail.

A useful way to state the target is to pair it with an action. If the action is selective answering, the target is the risk of an incorrect accepted answer. If the action is IDK, the target is evidence sufficiency or answerability. If the action is request-better-input, the target is perceptual adequacy. If the action is escalation, the target combines uncertainty with deployment risk. This action-paired view prevents the common mistake of using one generic confidence number for every decision.

### *5.2. Post-hoc, prompt-based, verbalized, and domain calibration*

Post-hoc calibration adjusts a score after prediction, for example through temperature scaling or related mapping functions. Prompt-based calibration changes the elicitation procedure so that the model reports more reliable confidence or chooses a safer response. Verbalized calibration focuses on aligning natural-language confidence with empirical correctness. Studies of MLLM calibration before and after multimodal training show that miscalibration can persist even when task performance improves [7]. Studies of verbalized confidence in VLMs show that confidence statements require evaluation rather than trust by default [53, 54].

Domain calibration is especially important when the application has specialized evidence and asymmetric error costs. Medical VQA studies evaluate confidence alignment in clinical visual reasoning and propose domain-aware calibration strategies [61, 62]. Clinical image-text work also studies how uncertainty propagates across image and language components in cardiac MR applications [63]. Affective visual reasoning work studies confidence verbalization and calibration in emotion understanding [64]. MLLM-as-judge settings also require calibration because the judge's score may vary across visual domains and prompt conditions [65]; related evaluator work shows that vision-language judges and implausibility evaluators should themselves be treated as uncertain systems [66]. The reporting rule is straightforward: calibration should be evaluated in the domain, modality, and action setting where the model will be used.

### *5.3. Source-aware and decoupled calibration*

A distinctive MLLM problem is source conflation. One confidence value may hide whether the system failed to see the object, failed to ground the claim, failed to reason, or faced an unanswerable question. Source-aware calibration tries to separate these cases. VL-Calibration explicitly separates visual confidence from reasoning confidence, so the model can distinguish perceptual uncertainty from inference uncertainty [9]. MMBoundary calibrates confidence at reasoning steps to improve knowledge-boundary awareness [8]. Grounding-based calibration uses visual support as part of the uncertainty estimate rather than treating confidence as a purely linguistic property [40].

This decomposition matters because different sources require different policies. Low visual confidence suggests asking for a better image or routing to a perception module. Low reasoning confidence suggests self-checking, sampling alternatives, or using a verifier. Low answerability suggests IDK or clarification. Calibration is therefore most useful when it preserves enough source information to guide action.

### *5.4. Conformal prediction and risk-control policies*

Conformal prediction and related risk-control methods move from calibrated scores to operational risk statements. They use a calibration set and a score function to construct



一个模型可以为了答案的正确性进行校准，但仍然无法传达答案是否有依据。因此，清晰的目标定义是方法的一部分，而不是报告细节。

一种有用的表述目标的方法是将其与一个动作配对。如果动作是选择性回答，目标就是错误被接受答案的风险。如果动作是“我不知道”，目标就是证据的充分性或可回答性。如果动作是请求更好的输入，目标就是感知的充分性。如果动作是升级，目标则是将不确定性与部署风险结合起来。这种动作配对的观点可以防止将每个决策都使用一个通用置信度数值的常见错误。

### *5.2.* 事后、基于提示、口头化及领域校准

事后校准是在预测之后调整分数，例如通过温度缩放或相关映射函数。基于提示的校准则改变提取程序，使模型报告更可靠的置信度或选择更安全的响应。口头化校准则关注将自然语言中的置信度与经验正确性对齐。在多模态训练前后对多任务大语言模型（MLLM）校准的研究表明，即使任务表现有所提升，校准错误仍可能存在 [7]。对视觉语言模型（VLM）中口头化置信度的研究表明，置信度陈述需要评估，而不能默认信任 [53, 54]。

域校准在应用中涉及专业证据和非对称错误成本时尤其重要。医学 VQA 研究评估临床视觉推理中的置信度对齐，并提出了面向领域的校准策略 [61, 62]。临床图像-文本工作还研究不确定性在心脏 MRI 应用中如何在图像和语言组件间传播 [63]。情感视觉推理工作研究情绪理解中的置信度表述和校准 [64]。多模态大语言模型作为评判者（MLLM-as-judge）的设置也需要校准，因为评判者的评分可能会因视觉领域和提示条件的不同而变化 [65]；相关的评估者工作表明，视觉-语言评判者和不合理性评估者本身也应被视为不确定系统 [66]。报告规则很简单：校准应在模型将被使用的领域、模态和操作环境中进行评估。

### *5.3.* 源感知与解耦校准

一个独特的多模态大语言模型（MLLM）问题是信息来源混淆。一个置信值可能掩盖系统是未能看到对象、未能验证声明、未能推理，还是面对一个无法回答的问题。基于来源的校准尝试区分这些情况。VL 校准明确将视觉置信与推理置信分开，因此模型可以区分感知不确定性与推理不确定性 [9]。MMBoundary 在推理步骤上校准置信，以提升知识边界意识 [8]。基于落地的校准使用视觉支持作为不确定性估计的一部分，而不是将置信度视为纯粹的语言属性 [40]。

这种分解很重要，因为不同的来源需要不同的策略。低视觉信心意味着需要请求更好的图像或将其传递给感知模块。低推理信心意味着需要自我检查、采样备选方案或使用验证器。低可回答性意味着‘不知道’或需要澄清。因此，当校准能够保留足够的来源信息以指导行动时，它最为有用。

### *5.4.* 保形预测与风险控制策略

保形预测及相关的风险控制方法从校准得分转向操作性风险声明。它们使用校准集和得分函数来构建

prediction sets, abstention thresholds, or decision rules with empirical coverage or risk-control properties under stated assumptions. TRON builds response sets for MLLMs and identifies high-quality answers with explicit risk-control targets [16]. Conformal abstention policies adapt thresholds for large language and vision-language models [17]. Split conformal prediction has been applied to VLM prediction sets in visual question answering [18]. Other work studies VLM uncertainty through conformal or uncertainty-aware evaluation lenses, including VLM evaluation, scene-graph generation, generative OCR, and report drafting [67, 68, 19, 20, 69].

The value of conformal and risk-control methods is not that they eliminate uncertainty. Their value is that they make the uncertainty policy explicit. Instead of saying that the model is uncertain, the system can say that the answer is accepted only when a target risk level is met, or that the output should be returned as a set, abstained from, or sent for review. The caveat is equally important: risk-control statements depend on the calibration distribution, the score function, and the assumed relation between calibration and deployment cases.

### *5.5. Calibration failure modes*

The main failure modes are overconfidence, underconfidence, source conflation, verbalization mismatch, and shift sensitivity. Overconfidence is dangerous because the model may provide unsupported answers in a tone of certainty. Underconfidence reduces usefulness through excessive abstention. Source conflation hides whether the problem is visual, linguistic, reasoning-related, or answerability-related. Verbalization mismatch appears when stated confidence does not track token-level or empirical confidence. Shift sensitivity appears when confidence calibrated in one domain, modality, prompt style, or video-frame setting fails elsewhere [70, 26, 36].

These failures motivate the next section. Calibration is not the end of the pipeline. Once uncertainty has been estimated and calibrated, the system must still decide what to do.

## 6. From Uncertainty to Action

Uncertainty-aware MLLMs should not answer every query with the same policy. The response should depend on the evidence condition, the calibrated risk, the user need, and the cost of error. This section treats action selection as part of the uncertainty problem itself. The action space includes answering, hedging, abstaining or saying IDK, clarifying, requesting better input, retrieving evidence, self-checking, routing, and escalation.

### *6.1. Answering, hedging, and selective answering*

Answering is appropriate when the available evidence is sufficient and the calibrated risk is acceptable. Hedging is appropriate when the evidence is partial but still useful, provided the hedge communicates a real limitation rather than disguising unsupported content. Selective answering formalizes this idea by allowing the model to answer only on cases where its expected risk is below a threshold. It is commonly evaluated through coverage, risk, and risk-coverage curves.

In visual question answering, selective methods ask whether a response should be accepted, abstained from, or clarified [12]. Perturbation, consistency, and variational uncertainty methods estimate when VLM responses are unreliable enough to reject or review [47, 13]. ReCoVERR reduces unnecessary abstention by gathering relevant



预测集、弃权阈值或在既定假设下具有经验覆盖率或风险控制特性的决策规则。TRON 为多标签大语言模型（MLLMs）构建响应集，并识别具有明确风险控制目标的高质量答案 [16]。符合性弃权策略为大语言模型和视觉-语言模型自适应阈值 [17]。分割符合性预测已应用于视觉问答中的视觉语言模型（VLM）预测集 [18]。其他工作通过符合性或不确定性评估视角研究视觉语言模型的不确定性，包括视觉语言模型评估、场景图生成、生成式OCR以及报告撰写 [67, 68, 19, 20, 69]。

保形和风险控制方法的价值不在于它们消除了不确定性。它们的价值在于将不确定性策略明确化。系统不再只是说模型存在不确定性，而是可以说明答案只有在达到目标风险水平时才被接受，或者输出应作为一个集合返回、放弃或送审。同样重要的警告是：风险控制声明依赖于校准分布、评分函数以及假设的校准与部署案例之间的关系。

### *5.5.* 校准故障模式

主要的失效模式有过度自信、缺乏自信、信息源混淆、表述不匹配以及对变化的敏感性。过度自信是危险的，因为模型可能以确定的语气提供无依据的答案。缺乏自信则通过过度避免回答降低了实用性。信息源混淆会掩盖问题是属于视觉、语言、推理还是可回答性相关。表述不匹配出现在声明的自信度与基于词元或经验的自信度不一致时。对变化的敏感性出现在在一个领域、模态、提示风格或视频帧设置中校准的自信度在其他环境中失效时 [70, 26, 36]。

这些失败促使下一部分的展开。校准并不是流程的终点。一旦不确定性被估计并校准，系统仍然必须决定下一步的行动。

## 6. 从不确定到行动

考虑不确定性的多模态大模型（MLLMs）不应对每个查询都使用相同的策略。响应应取决于证据状况、校准风险、用户需求以及错误成本。本节将动作选择视为不确定性问题本身的一部分。动作空间包括回答、保留态度、回避或说‘我不知道’、澄清、请求更好的输入、检索证据、自我检查、路由和升级。

### *6.1.* 回答、回避和选择性回答

当可用证据充分且校准后的风险可接受时，回答是适当的。当证据部分但仍有用时，对冲是适当的，前提是对冲传达了真实的限制，而不是掩盖无支持的内容。选择性回答通过允许模型仅在其预期风险低于阈值的情况下回答，从而形式化了这一思想。通常通过覆盖率、风险以及风险-覆盖率曲线进行评估。

在视觉问答中，选择性方法会询问是否应该接受、回避或澄清一个回答 [12]。扰动、一致性和变分不确定性方法用于估计 VLM 的回答何时不够可靠以至于需要被拒绝或审查 [47, 13]。ReCoVERR 通过收集相关信息来减少不必要的回避

and reliable visual evidence before refusing to answer [71]. Variational VQA frames uncertainty-aware selective prediction through posterior uncertainty and reports benefits under distribution shift [13]. Audio-visual QA work extends selective prediction to settings where both acoustic and visual evidence matter [29]. The shared lesson is that abstention should be neither automatic nor stigmatized; it should be a calibrated decision.

### *6.2. IDK, refusal, and answerability*

IDK behavior is most useful when it is tied to answerability. A model should not say "I do not know" merely because a probability is low; it should say so when the input does not contain enough evidence, when the evidence is incompatible with the question, or when the required fact lies outside the supplied context. Unsolvable problem detection studies absent-answer, incompatible-answer-set, and incompatible-visual-question cases [22]. MM-SAP evaluates whether MLLMs recognize known and unknown perception cases [14]. Visual-IDK and MoHoBench evaluate knowledge boundaries and honesty on visually unanswerable questions [15, 24]. Video and multimodal reasoning benchmarks extend the same idea to temporal and evidence-sufficiency settings [23, 25].

Answerability also matters in embodied settings. An agent may lack the relevant view, face an underspecified instruction, or need information from a user. AbstainEQA studies embodied question answering cases where the agent should abstain because of missing context, underspecification, preference dependence, information unavailability, or false presuppositions [28]. These settings make clear that IDK is not just a language behavior; it is an interaction policy under partial evidence.

### *6.3. Clarification and better-input requests*

Clarification and better-input requests are different actions. Clarification is appropriate when the user question is ambiguous: the object reference is unclear, several interpretations are plausible, or the user's goal is underspecified. Better-input requests are appropriate when the question may be clear but the sensory evidence is not: the image is blurred, the document crop is incomplete, the audio is noisy, or the video omits the relevant moment.

This distinction is central for multimodal systems. ClearVQA trains and evaluates VLMs that ask clarification questions for ambiguous visual questions rather than forcing a direct answer [72]. Right This Way studies whether VLMs can guide a user to obtain better visual evidence when the current view is insufficient [39]. Embodied help-seeking systems study how agents can ask minimal clarification questions under partial observability [73]. These actions preserve usefulness: the model does not simply refuse, but tries to obtain the missing condition for a reliable answer.

### *6.4. Retrieval, self-checking, and verification*

Retrieval is appropriate when the input is visually sufficient but external knowledge is needed. Self-checking and verification are appropriate when the answer may be visually grounded but the reasoning path or factual claim is unstable. Agentic frameworks use uncertainty to decide when to call a tool, ask a verifier, compare candidates, or refine an answer.

MMA-RAG uses internal multimodal representations to decide when retrieval should be used for VQA [59]. SRICE uses uncertainty estimates to guide external tool use and answer selection in multimodal reasoning [74]. Multi-agent VQA calibration uses interacting agents to critique, refine, and aggregate responses [57]. Consensus-style OCR systems use agreement among multiple VLMs to identify low-quality outputs and support



在拒绝回答 [71] 之前，需要可靠的视觉证据。变分 VQA 通过后验不确定性来构建考虑不确定性的选择性预测，并报告了分布转移下的优势 [13]。视听问答工作将选择性预测扩展到声学和视觉证据都重要的场景 [29]。共同的经验是，弃权既不应自动发生，也不应被污名化；它应当是经过校准的决策。

### *6.2.* 不知道、拒绝与可回答性

当“我不知道”的行为与可回答性相关时，它最为有用。模型不应仅因为某个概率较低就说“我不知道”；它应该在输入信息不足、证据与问题不兼容，或所需事实超出提供的上下文时才这样说。不可解问题检测研究缺失答案、不兼容答案集以及不兼容视觉问题的情况[22]。MM-SAP评估多模态大语言模型是否能识别已知和未知的感知案例[14]。Visual-IDK和MoHoBench评估在视觉上不可回答的问题上的知识边界和诚实性 [15, 24]。视频和多模态推理基准将同样的理念扩展到时间和证据充分性场景中 [23, 25]。

可回答性在具身环境中也很重要。智能体可能缺乏相关视角、面对不完整的指令，或需要来自用户的信息。AbstainEQA研究了具身问答场景，其中智能体应因缺失的上下文、不完全说明、依赖偏好、信息不可用或错误的前设而选择不回答 [28]。这些情境清楚地表明，‘我不知道’不仅仅是一种语言行为；它是在部分证据下的一种交互策略。

### *6.3.* 澄清与更佳输入请求

澄清和更好的输入请求是不同的操作。当用户的问题不明确时，适合进行澄清：对象引用不清楚、存在多种可能的解释，或者用户的目标没有明确说明。当问题可能清楚但感官证据不充分时，适合提出更好的输入请求：图像模糊、文档裁剪不完整、音频嘈杂，或者视频遗漏了相关时刻。

这种区分对于多模态系统至关重要。ClearVQA 训练和评估能够针对模糊视觉问题提出澄清问题的视觉语言模型，而不是强行给出直接答案 [72]。Right This Way 研究视觉语言模型是否能够在当前视图不足时，引导用户获取更好的视觉证据 [39]。具身求助系统研究代理如何在部分可观察情况下提出最少的澄清问题 [73]。这些行为保持了实用性：模型不会简单地拒绝，而是尝试获取可靠答案所缺失的条件。

### *6.4.* 提取、自我检查与核实

当输入在视觉上足够但需要外部知识时，检索是适当的。当答案可能有视觉支撑但推理路径或事实声明不稳定时，自我检查和验证是适当的。自主框架利用不确定性来决定何时调用工具、询问验证者、比较候选项或完善答案。

MMA-RAG 使用内部多模态表示来决定什么时候应该在 VQA 中使用检索 [59]。SRICE 使用不确定性估计来指导外部工具的使用和多模态推理中的答案选择 [74]。多代理 VQA 校准使用相互作用的代理来批评、完善和聚合回答 [57]。共识式 OCR 系统使用多个 VLM 之间的一致性来识别低质量输出并提供支持

self-verification [58]. Clinical hallucination detection uses repeated reports and calibrated thresholds to flag uncertain findings for review [51]. These methods show that uncertainty can be used not only to stop an answer, but also to improve the evidence pathway before answering.

### *6.5. Routing and escalation*

Routing and escalation are needed when uncertainty is high and the cost of error is unacceptable. Routing may send a case to a stronger model, a specialized model, an external tool, or a cheaper model when the case is easy. Escalation may send the case to a human expert or domain workflow. Adaptive VLM routing for computer-use agents estimates difficulty and confidence, then routes actions to a model that satisfies a target reliability threshold [60]. In medical, clinical, robotic, or high-stakes settings, escalation is not merely an engineering convenience; it is part of the safety policy.

Routing also prevents a false choice between full autonomy and full refusal. A model can answer low-risk cases, retrieve or self-check medium-risk cases, and escalate high-risk cases. This tiered policy is often more useful than a binary answer/abstain rule, especially when uncertainty comes from several sources at once.

### *6.6. Source-action alignment*

Table 4 summarizes the main action policies. The table is intentionally qualitative: count-heavy source-action diagnostics are retained for traceability, while the main text emphasizes the decision logic. The same uncertainty score should not trigger the same action in every source condition.



自我验证 [58]。临床幻觉检测使用重复报告和校准阈值来标记不确定的发现以供审查 [51]。这些方法表明，不确定性不仅可以用来停止给出答案，还可以在回答前改善证据路径。

### *6.5.* 路由与升级

当不确定性高且错误成本无法接受时，需要进行路由和升级。路由可能会将案例发送到更强的模型、专门的模型、外部工具，或者在案例简单时发送到更便宜的模型。升级可能会将案例发送给人工专家或特定领域的工作流程。用于计算机使用代理的自适应 VLM 路由会估计难度和置信度，然后将操作路由到满足目标可靠性阈值的模型。在医疗、临床、机器人或高风险场景中，升级不仅仅是工程上的便利，它是安全策略的一部分。

路由还可以防止在完全自主和完全拒绝之间做出错误选择。模型可以回答低风险的情况，检索或自我检查中等风险的情况，并升级高风险的情况。这种分层策略通常比二元的回答/放弃规则更有用，特别是当不确定性同时来自多个来源时。

### *6.6.* 源-动作对齐

表4总结了主要的行动策略。该表格故意采用定性方式：保留了以计数为主的来源-行动诊断以便追踪，而正文则强调决策逻辑。同样的不确定性分数不应在每种来源条件下触发相同的行动。

Table 4: Action-policy taxonomy for uncertainty-aware MLLMs. The table maps evidence conditions and uncertainty signals to response actions. Citations attached to action labels are representative examples; the trigger and failure-mode columns are synthesized policy implications.

| **Action** | **When it is appropriate** | **Signals that can trigger it** | **Potential failure mode** |
|---|---|---|---|
| Answer [12, 9, 40] | Evidence is sufficient and calibrated risk is within the accepted range. | High grounding support, stable samples, calibrated confidence, low shift signal. | Confident hallucination if the threshold is too loose. |
| Hedge [53, 54, 64] | Evidence is partial but still useful, and the limitation can be stated precisely. | Moderate confidence, mild disagreement, incomplete but relevant support. | Vague caution that hides a lack of evidence. |
| Abstain or IDK [22, 14, 15, 24, 23] | The required evidence is missing, incompatible, outside the input, or beyond the model boundary. | Answerability score, IDK probability, large prediction set, unsupported-evidence flag. | Excessive refusal and reduced task usefulness. |
| Clarify [72, 73, 28] | The user's question, referent, intent, or preference is underspecified. | Ambiguous reference, multiple plausible interpretations, conflicting user context. | Unnecessary interaction when the answer was already clear enough. |
| Request better input [39, 73, 28] | The query is clear but the sensory evidence is degraded or incomplete. | Blur, occlusion, low OCR confidence, visual perturbation instability, missing video segment. | Burdening the user when the model should use available evidence. |
| Retrieve, self-check, or verify [59, 74, 57, 58, 51] | External knowledge, evidence checking, or reasoning validation is needed before answering. | Reasoning disagreement, weak factual support, verifier uncertainty, tool-need signal. | Added latency, retrieval noise, or over-reliance on an imperfect verifier. |
| Route or escalate [60, 61, 28] | Deployment risk is high, model competence is insufficient, or the case violates the autonomous policy boundary. | High calibrated risk, domain shift, safety-critical context, expert-domain flag. | Excessive cost or human burden if thresholds are too conservative. |

The action taxonomy completes the source–signal–calibration–action chain. Evaluation should therefore test not only whether a score predicts errors, but whether the selected action is appropriate for the evidence condition and deployment risk.

## 7. Evaluating Uncertainty-Aware Behavior

Evaluation is where uncertainty-aware MLLM research becomes comparable. A method may produce an apparently useful confidence score, but its practical meaning depends on the benchmark, the correctness target, the uncertainty source being tested, and the action the score is meant to support. For this reason, evaluation should not stop at final-answer accuracy or at a single calibration number. It should ask whether the model's uncertainty improves behavior when the evidence is insufficient, conflicting, shifted, or high-risk.



表4：面向不确定性感知多模态大模型的行动-策略分类法。该表将证据条件和不确定性信号映射到响应动作。附加在动作标签上的引用是代表性示例；触发和失败模式列是综合的策略含义。

| **动作** | **当它是适当的** | **可以的信号触发它** | **潜在故障 mode** |
|---|---|---|---|
| 回答 [12, 9, 40] | 证据充分以及校准风险是在可接受范围内范围。 | 高地放电支撑，稳定的样本，校准后的信心低档信号。 | 自信的幻觉如果阈值太松散。 |
| 对冲 [53, 54, 64] | 证据是部分的，但仍然有用，而且可以陈述限制精确地。 | 适度信心，轻微分歧，不完整但相关支持 | 含糊的警告，指出掩盖了缺乏证据。 |
| 弃权或不知道 [22, 14, 15, 24, 23] | 所需的证据丢失，不兼容，外部输入，或超出模型边界。 | 可回答性得分，我不知道概率，很大预测集不支持的证据旗. | 过度拒绝和任务有用性降低。 |
| 澄清 [72, 73, 28] | 用户的问题，指称、意图，或偏好是未具体说明的 | 含糊不清的引用，多种可能解释，冲突的用户上下文。 | 不必要的互动当答案是已经足够清楚了。 |
| 请求更好输入 [39, 73, 28] | 问题很清楚，但是感官证据是降解或不完整。 | 模糊，遮挡，低OCR 置信度，视觉扰动不稳定性缺失的视频片段。 | 给用户带来负担当模型应该使用可用的证据。 |
| 取回，自我检查，或验证 [59, 74, 57, 58, 51] | 外部知识，证据核查，或者推理验证是之前需要回答中。 | 推理分歧，弱事实支持，验证者不确定性，工具需求信号。 | 增加的延迟，检索噪音，或过度依赖在一个不完善的验证器上。 |
| 路线或升级 [60, 61, 28] | 部署风险很高，模型能力是不足，或者情况违反自治政策边界。 | 高校准风险，领域转移安全关键环境，专家领域标志。 | 过高的成本或人类负担如果阈值也太保守的。 |

动作分类法完成了源–信号–校准–动作链。因此，评估不仅应测试分数是否能够预测错误，还应测试所选择的动作是否适合证据条件和部署风险。

## 7. 评估不确定性感知的行为

评估是使具有不确定性感知的多模态大语言模型（MLLM）研究具有可比性的环节。一个方法可能产生看似有用的置信度分数，但其实际意义取决于基准、正确性的目标、所测试的不确定性来源以及该分数旨在支持的操作。因此，评估不应仅停留在最终答案的准确性或单一的校准数值上。评估应当关注模型的不确定性在证据不足、冲突、变化或高风险的情况下是否能改善行为。

A useful evaluation protocol therefore needs to connect four elements. First, the benchmark must specify the task, the evidence-condition label or controlled evidence construction, and the relevant risk context. Second, the method must state which signal is being used and what model access it requires. Third, calibration or thresholding must convert the signal into a decision rule. Fourth, the evaluation must score both answer quality and action quality. Figure 5 summarizes this evaluation view. The figure is intentionally schematic: its purpose is to show what should be reported, not to encode corpus counts.

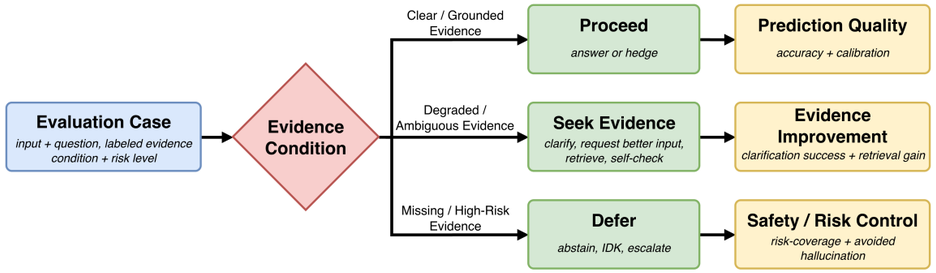



Figure 5: **Evaluating uncertainty-aware behavior.** Evaluation should test whether the model's uncertainty leads to an appropriate action for a labeled or controlled evidence condition. Clear and grounded evidence should support proceeding with an answer, degraded or ambiguous evidence should trigger evidence-seeking actions such as clarification, better-input requests, retrieval, or self-checking, and missing evidence or high-risk contexts should support deferral through abstention, IDK, or escalation. The evaluation target therefore depends on the selected action: prediction quality for proceeding, evidence improvement for seeking additional information, and safety or risk control for deferral. When evidence-condition labels are unavailable, reliable source- and action-aware evaluation remains an open benchmark challenge.

In this protocol, the evidence condition is not assumed to be obvious from the model output. In controlled evaluations, it can be specified by benchmark construction, for example through blur, crop, occlusion, distractors, missing-evidence prompts, text–image conflict, or high-risk domain labels. In naturalistic evaluations, it requires annotation or metadata such as object visibility, answerability, grounding support, reference evidence, or expert judgment. When such labels are absent, the evaluation can still measure correlations between uncertainty and error, but it cannot fully test whether the selected action matched the underlying source of uncertainty.

### *7.1. Benchmark families*

Benchmarks for uncertainty-aware MLLMs can be grouped by the behavior they test rather than by dataset name alone. General multimodal QA and reasoning benchmarks are useful for measuring the relationship between confidence and correctness under ordinary visual and reasoning demands. However, they often provide only post-hoc correctness labels, so they cannot always explain whether an error came from perception, grounding, reasoning, or answerability.

A second family focuses on unanswerability, refusal, and IDK behavior. These settings are especially important for MLLMs because a visual question can be unanswerable even when the language prompt is well formed: the relevant object may be outside the image, hidden, too small, temporally absent from a video, contradicted by the prompt, or unavailable without external knowledge. Work on selective VQA, multimodal



因此，一个有用的评估方案需要连接四个要素。首先，基准必须指定任务、证据条件标签或受控证据构建，以及相关的风险背景。其次，方法必须说明使用了哪种信号以及所需的模型访问权限。第三，校准或阈值处理必须将信号转换为决策规则。第四，评估必须对答案质量和行动质量进行评分。图5总结了这一评估视图。该图特意采用示意方式：其目的是展示应报告的内容，而不是编码语料数量。

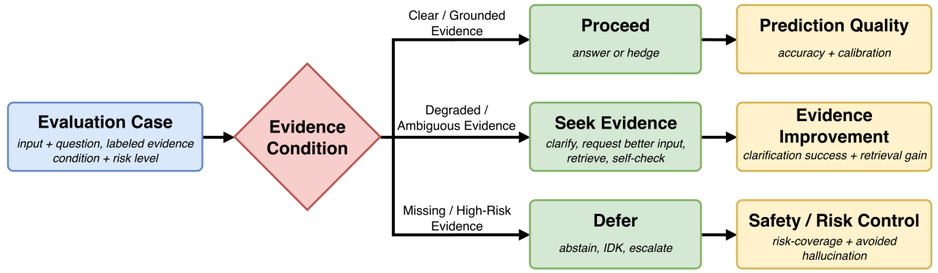



图5：评估不确定性感知行为。评估应测试模型的不确定性是否会导致对有标签或可控证据条件采取适当的行动。明确且有依据的证据应支持继续给出答案，退化或模糊的证据应触发寻求证据的行为，如澄清、更好的输入请求、检索或自检，而缺失的证据或高风险情境应支持通过弃权、不知道或升级来推迟决策。因此，评估目标取决于所选择的行动：对继续的评估是预测质量，对寻求额外信息的评估是证据改进，对推迟的评估是安全性或风险控制。当证据条件标签不可用时，可靠的源和行动感知评估仍然是一个开放的基准挑战。

在本协议中，不假定证据条件可以从模型输出中显而易见。在受控评估中，可以通过基准构建来指定，例如通过模糊、裁剪、遮挡、干扰项、缺失证据的提示、文本–图像冲突或高风险领域标签。在自然评估中，则需要注释或元数据，如物体可见性、可回答性、基础支持、参考证据或专家判断。当这些标签缺失时，评估仍可以测量不确定性与错误之间的相关性，但无法完全测试所选择的行动是否与潜在的不确定性来源相匹配。

### *7.1.* 基准系列

针对不确定性感知多模态大语言模型（MLLM）的基准测试可以根据它们所测试的行为来分类，而不仅仅是根据数据集名称。通用多模态问答和推理基准测试有助于衡量在常规视觉和推理需求下的置信度与正确性之间的关系。然而，它们通常只提供事后的正确性标签，因此无法总是解释错误是来自感知、落地、推理还是可答性。

第二类关注于不可回答性、拒绝和“我不知道”的行为。这些设置对于多语言大模型特别重要，因为即使语言提示格式良好，视觉问题也可能无法回答：相关对象可能在图像之外、被遮挡、太小、在视频中暂时不存在、与提示相矛盾，或者没有外部知识无法获取。在选择性视觉问答、多模态方面的工作

self-awareness, visual IDK behavior, multimodal honesty, video refusal, embodied QA, audio-visual QA, and chart-focused evaluation shows why the correct action is sometimes not to answer [12, 14, 15, 24, 23, 28, 29, 75].

A third family targets hallucination and grounding. These benchmarks ask whether the generated answer is supported by the supplied evidence. Object-hallucination and groundedness metrics are useful for detecting unsupported content, but they should not be treated as a complete substitute for calibration. A method can reduce hallucination without producing a reliable confidence estimate, and a confidence estimate can detect likely errors without telling the user whether the failure was visual, factual, or inferential [47, 37, 48, 76, 40].

A fourth family evaluates explicit risk control. Conformal and selective-prediction benchmarks ask whether the system can satisfy a target risk or coverage level, sometimes by returning a prediction set, abstaining, or selecting among candidate responses. These protocols are valuable because they turn uncertainty into a measurable operating point rather than a descriptive score [16, 17, 18, 71, 67].

The final family is domain and modality specific. Medical images, clinical summaries, audio, video, charts, OCR, documents, scientific inputs, embodied observations, and omni-modal settings expose failure modes that generic image-text benchmarks may miss. In these domains, evaluation should report whether uncertainty is calibrated in the target context, not only whether it transfers from a general benchmark [61, 62, 50, 69, 63, 25, 38, 27, 41, 21, 30].

### *7.2. Metric families*

Metric choice should follow the evaluation question. Calibration metrics, such as expected calibration error, Brier score, negative log-likelihood, and reliability diagrams, ask whether a reported confidence value matches empirical correctness for a defined target. They are necessary, but the target must be stated carefully. Confidence may refer to final-answer correctness, grounding support, answerability, a reasoning step, a verifier score, or a risk-controlled action. A model can be calibrated for multiple-choice accuracy while still being poorly calibrated about whether its answer is visually grounded.

Detection metrics treat uncertainty as a ranking signal. AUROC, AUPRC, F1, error-detection accuracy, and related screening measures ask whether uncertain outputs are more likely to be wrong or unsupported. These metrics are useful for filtering and triage, but they do not by themselves define what the model should do after detecting risk. Selective-prediction metrics add the action layer by measuring coverage, risk, risk-coverage curves, abstention precision, area-under-curve variants, and response quality on the answered subset.

Conformal and risk-control metrics make the operating target explicit. They measure empirical coverage, violation rate, prediction-set size, set efficiency, or risk at a specified level. These metrics are attractive when a system must satisfy a defined safety or coverage constraint, but they depend on the calibration data and may degrade under shift. Answerability and honesty metrics ask a different question: whether the model knows when the supplied evidence is enough. Grounding and evidence-support metrics ask whether the answer is anchored to the correct input evidence. Human-facing metrics ask whether uncertainty communication helps users make better decisions.

### *7.3. Evaluation protocol checklist*

Table 5 reorganizes metrics around the question they answer. The main lesson is that no single metric is sufficient. A strong evaluation should report the uncertainty source,



自我意识、视觉不确定行为、多模态诚实、视频拒绝、具身问答、视听问答以及以图表为中心的评估显示了为什么正确的行为有时是不回答 [12, 14, 15, 24, 23, 28, 29, 75]。

第三类评估针对幻觉和依据性。这些基准测试询问生成的答案是否有提供的证据支持。对象幻觉和依据性指标对于检测不支持的内容很有用，但不应被视为校准的完整替代方法。一种方法可以减少幻觉，但未必能生成可靠的置信度估计，而置信度估计可以检测可能的错误，却无法告诉用户失败是视觉的、事实的，还是推理的。

第四类方法评估显式的风险控制。保形和选择性预测基准测试系统是否能够满足目标风险或覆盖水平，有时通过返回预测集、回避决策或在候选响应中进行选择。这些协议的价值在于，它们将不确定性转化为可测量的操作点，而不是描述性分数。

最后一类是特定于领域和模态的。医学影像、临床摘要、音频、视频、图表、OCR、文档、科学输入、具体观察以及全模态设置揭示了通用图文基准可能遗漏的失败模式。在这些领域中，评估应报告不确定性在目标环境中是否得到校准，而不仅仅是它是否可以从通用基准中迁移。

### *7.2.* 指标系列

指标选择应根据评估问题进行。校准指标，例如期望校准误差、布里尔分数、负对数似然和可靠性图，用于判断报告的置信度值是否与定义目标的实际正确性相符。这些指标是必要的，但必须明确说明目标。置信度可能指最终答案的正确性、基础支持、可回答性、推理步骤、验证器得分或风险控制操作。一个模型在多选题准确性上可以是校准良好的，但在其答案是否有视觉依据方面仍可能校准不良。

检测指标将不确定性视为排序信号。AUROC、AUPRC、F1、错误检测准确率及相关筛查指标会评估不确定的输出是否更可能是错误的或不支持的。这些指标对于过滤和分流非常有用，但它们本身并不定义模型在检测到风险后应该做什么。选择性预测指标通过测量覆盖率、风险、风险-覆盖曲线、弃权精度、曲线下面积变体以及已回答子集的响应质量，增加了行动层的考量。

保形和风险控制指标使操作目标明确。它们衡量经验覆盖率、违规率、预测集大小、集效率或指定水平的风险。当系统必须满足定义的安全或覆盖约束时，这些指标很有吸引力，但它们依赖于校准数据，并且在数据变化时可能会退化。可回答性和诚实度指标提出了不同的问题：模型是否知道提供的证据是否足够。基础性和证据支持指标则问答案是否依赖于正确的输入证据。面向人类的指标则问不确定性沟通是否有助于用户做出更好的决策。

### *7.3.* 评估协议清单

表5围绕它们所回答的问题重新组织了指标。主要的教训是，没有单一指标是足够的。一个强有力的评估应该报告不确定性的来源，

model-access assumption, signal family, calibration procedure, action policy, and action cost. This is particularly important for black-box MLLMs, where repeated sampling, verbalized confidence, and judge scores may be the only accessible uncertainty signals.

Table 5: Evaluation guide for uncertainty-aware MLLMs. The table links evaluation questions to benchmark families, metric families, and hidden failure modes. Citations in the benchmark-family column are representative examples of the evaluation setting.

| **Evaluation question** | **Benchmark family** | **Metric family** | **What can remain hidden** |
|---|---|---|---|
| Is confidence aligned with correctness? | General VQA, chart QA, medical VQA, multimodal reasoning, open-ended QA [7, 9, 61, 62, 53]. | ECE, Brier score, NLL, reliability diagrams, confidence-accuracy gap. | Whether the answer is grounded, answerable, or safe to act on. |
| Can uncertainty rank risky outputs? | Error detection, hallucination detection, selective prediction, verifier screening [12, 47, 13, 51, 36]. | AUROC, AUPRC, F1, TPR at fixed FPR, risk-coverage curves. | Which action should follow after a risky output is detected. |
| Does the model know when evidence is insufficient? | Unanswerable VQA, visual IDK, video refusal, embodied QA, audio-visual QA [22, 14, 15, 24, 23, 29]. | Answerability accuracy, abstention precision, refusal helpfulness, IDK calibration. | Whether abstention is too conservative or insufficiently informative. |
| Can the system control risk? | Conformal MLLM/VLM benchmarks, prediction-set evaluation, risk-controlled response selection [16, 17, 18, 19, 20]. | Empirical coverage, violation rate, prediction-set size, set efficiency, controlled risk. | Robustness under domain shift, prompt shift, and calibration-data mismatch. |
| Does uncertainty improve user-facing behavior? | Grounded QA, clinical or document workflows, interactive agents, high-risk settings [71, 72, 39, 59, 74]. | Action utility, escalation appropriateness, human ratings, explanation quality, support precision. | Whether users understand the uncertainty source and can act on it safely. |

The reporting checklist should be explicit. Each study should identify the base model and version, modality, benchmark, prompt format, model-access level, uncertainty signal, calibration data, target variable, thresholds, action policy, action costs, and failure cases. Without these details, two methods may appear comparable while actually measuring different targets.

Evaluation choices become especially important once a model leaves a benchmark setting. The next section therefore asks how modality, domain risk, and user interaction change the action policy attached to uncertainty.



模型访问假设、信号家族、校准程序、动作策略和动作成本。这对于黑箱多模态大语言模型尤其重要，在这些模型中，重复采样、口头化的置信度和评判分数可能是唯一可以获取的不确定性信号。

表 5：不确定性感知多模态大模型的评估指南。该表将评估问题与基准系列、指标系列和潜在失败模式关联起来。基准系列列中的引用是评估环境的代表性示例。

| **评估问题** | **基准系列** | **公制系列** | **什么可以留下隐藏** |
|---|---|---|---|
| 是自信与……对齐正确性? | 通用视觉问答，图表问答，医学视觉问答，多模态推理开放式问答 [7, 9, 61, 62, 53]。 | ECE，Brier 分数，NLL，可靠性图，信心-准确性 gap. | 无论答案是有根据的，可回答的，或者安全可行。 |
| 可能的不确定性排名有风险输出? | 错误检测幻觉检测，选择性预测验证者筛选 [12, 47, 13, 51, 36]。 | AUROC，AUPRC，F1，在固定假阳性率下的真正率，风险覆盖曲线。 | 应该采取哪种行动跟随一个危险的人检测到输出。 |
| 模型是否知道什么时候证据是不足? | 无法回答的视觉问答视觉不知道，视频拒绝，体现的质量保证视听问答 [22, 14, 15, 两千四百二十三 29]。 | 可回答性准确性，弃权精确，拒绝乐于助人，不知道校准。 | 是否弃权过于保守或不足地信息丰富。 |
| 系统可以控制风险? | 共形多模态大语言模型/视觉语言模型基准，预测集评估，风险可控响应选择 [16, 17, 18, 19, 20]。 | 经验覆盖率违规率预测集大小，集合效率，受控风险。 | 在……下的稳健性领域转移，提示转变，以及校准数据不匹配。 |
| 不确定性改善面向用户行为? | 扎根问答，临床或文件工作流程，互动代理，高风险设置 [71, 72, 39, 59, 74]。 | 行动效用，升级适当性，人工评分，解释质量，支持精确性。 | 无论用户理解这个不确定性来源和可以安全地采取行动。 |

报告清单应该明确。每个研究都应明确基础模型及其版本、模态、基准、提示格式、模型访问级别、不确定性信号、校准数据、目标变量、阈值、行动策略、行动成本和失败案例。没有这些细节，两种方法可能看起来可比，但实际上测量的是不同的目标。

一旦模型离开基准测试环境，评估选择就变得尤为重要。因此，下一节将讨论模态、领域风险和用户交互如何改变与不确定性相关的行动策略。

## 8. Deployment Landscape

Applications determine which uncertainty sources matter and which actions are acceptable. In a low-risk visual QA benchmark, abstention may simply reduce coverage. In medical, embodied, accessibility, or safety-critical settings, the same abstention may prevent harm. Deployment therefore changes the meaning of uncertainty. The same signal can support different policies depending on the cost of a wrong answer, the cost of delay, the availability of human review, and the possibility of collecting better evidence.

### *8.1. High-risk and expert-mediated domains*

Medical, clinical, embodied, and safety-critical applications require conservative uncertainty policies. In these settings, calibration should be evaluated in the target domain, grounding should be inspectable, and escalation should be part of the workflow rather than an afterthought. Clinical and medical studies highlight confidence calibration, uncertainty propagation, grounded evidence, and risk-controlled reporting because fluent but unsupported responses can be harmful [61, 62, 63, 50, 69]. Embodied and robot-facing settings add another layer: uncertainty may need to stop an action, request a new observation, or ask a human for confirmation rather than merely change a textual answer [28].

Performance-prediction and routing work also treats uncertainty as an input to deployment decisions across vision-language tasks [77]. A useful design principle is to match the policy to the deployment risk. Low-risk tasks can tolerate a wider answer region and use uncertainty for hedging or selective answering. High-risk tasks need stricter thresholds, audit trails, and escalation routes. This does not mean that high-risk systems should always refuse; over-refusal can also reduce usefulness. It means the policy should make the cost of errors, abstentions, retrieval, and escalation explicit.

### *8.2. Modality-specific uncertainty*

Different modalities create different uncertainty profiles. Static image-text QA emphasizes visual ambiguity, perception, object grounding, and language-prior dominance. Video adds temporal answerability: the relevant event may not occur in the observed clip, may occur outside the sampled frames, or may require temporal ordering. Audio adds acoustic quality, speaker ambiguity, and cross-modal timing. Charts and OCR introduce symbolic extraction, layout, scale, and numerical uncertainty. Documents introduce evidence localization across text, tables, and figures. Omni-modal and multimodal dissonance settings add the possibility that modalities disagree rather than merely complement one another [23, 25, 75, 38, 27, 41, 30, 21].

This modality view helps explain why a universal confidence score is often too weak. A video model may need to say that the event is not visible in the clip. A chart model may need to say that the axis cannot be read reliably. A document model may need to cite the page or table that supports the answer. An audio-visual model may need to report that the audio and visual evidence conflict. These are not merely different benchmarks; they are different uncertainty-action patterns.

### *8.3. Human-facing uncertainty communication*

Uncertainty-aware deployment is also a communication problem. Users do not only need to know that a model is uncertain; they need to know why and what to do next. A vague hedge can hide the absence of evidence. A refusal can be correct but unhelpful if it does not explain what information is missing. A confidence score can be misleading if



## 8. 部署环境

应用程序决定哪些不确定性来源重要，以及哪些操作是可接受的。在低风险的视觉质量评估基准中，回避可能只是减少覆盖率。在医疗、具身、无障碍或安全关键的环境中，相同的回避可能会防止伤害。因此，部署会改变不确定性的含义。相同的信号可以根据错误答案的代价、延迟的代价、人类审查的可用性以及收集更好证据的可能性支撑不同的策略。

### *8.1.* 高风险和专家介入领域

医疗、临床、具身化和安全关键的应用需要保守的不确定性策略。在这些环境中，应在目标领域评估校准，确保基础信息可检查，并且升级机制应作为工作流程的一部分而非事后补充。临床和医疗研究强调置信度校准、不确定性传播、基于证据的依据以及风险可控的报告，因为流利但缺乏支持的回答可能有害 [61, 62, 63, 50, 69]。具身化和面向机器人的场景增加了另一层要求：不确定性可能需要停止一个动作、请求新的观察或向人类确认，而不仅仅是改变文本答案 [28]。

性能预测和路由工作也将不确定性作为输入，用于在视觉语言任务中做出部署决策 [77]。一个有用的设计原则是将策略与部署风险相匹配。低风险任务可以容忍更广的答案范围，并利用不确定性进行对冲或选择性回答。高风险任务则需要更严格的阈值、审计跟踪和升级路径。这并不意味着高风险系统应始终拒绝；过度拒绝也会降低实用性。这意味着策略应明确错误、弃权、检索和升级的成本。

### *8.2.* 模态特定的不确定性

不同的模态会产生不同的不确定性特征。静态图像-文本问答强调视觉模糊性、感知、对象定位以及语言先验的主导性。视频增加了时间上的可回答性：相关事件可能未在观察到的片段中发生，可能发生在抽取帧之外，或者可能需要时间顺序。音频增加了声学质量、说话人模糊性以及跨模态的时序。图表和OCR引入了符号提取、布局、尺度和数值不确定性。文档引入了跨文本、表格和图形的证据定位。全模态和多模态不一致设置增加了模态可能存在分歧而不仅仅是互补的可能性。

这种模态视角有助于解释为什么一个通用的置信度评分往往过于薄弱。视频模型可能需要说明事件在剪辑中不可见。图表模型可能需要说明坐标轴无法被可靠读取。文档模型可能需要引用支持答案的页面或表格。视听模型可能需要报告音频和视觉证据存在冲突。这些不仅仅是不同的基准测试；它们是不同的不确定性-行动模式。

### *8.3* 面向人的不确定性沟通

考虑不确定性的部署也是一个沟通问题。用户不仅需要知道模型存在不确定性；他们还需要知道原因以及接下来该做什么。模糊的保留措辞可能会掩盖证据的缺失。拒绝可能是正确的，但如果没有解释缺失的信息，则无益。置信度评分可能会产生误导，如果

users do not know whether it reflects visual grounding, answerability, or language-model fluency.

For human-facing systems, uncertainty communication should be specific and action-oriented. The model should distinguish between weak input quality, missing evidence, conflicting modalities, unstable reasoning, and high deployment risk. It should also give the user a useful next step: upload a clearer image, specify the target object, provide the missing context, allow retrieval, consult an expert, or treat the answer as tentative. This framing connects uncertainty estimation to trust calibration and accountability.

These deployment pressures expose the main gaps in the field: methods must become more source-aware, more action-aware, more robust under shift, and easier to reproduce across models and modalities.

## 9. Open Problems and Research Agenda

The research agenda follows directly from the source–signal–calibration–action chain. The near-term priority is to make uncertainty evaluation easier to compare. The mid-term priority is to make uncertainty estimates source-aware and robust under realistic access constraints. The longer-term priority is to integrate uncertainty into interactive, tool-using, and human-facing multimodal systems. Figure 6 summarizes these priorities as a time-horizon roadmap.

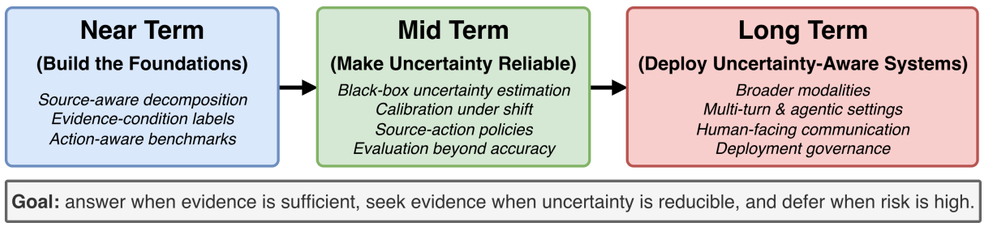



Figure 6: **Research roadmap for uncertainty-aware MLLMs.** The roadmap organizes future work by time horizon. Near-term work should build the foundations for source-aware uncertainty analysis, including evidence-condition labels and action-aware benchmarks. Mid-term work should make uncertainty estimates reliable under realistic access and deployment conditions through black-box uncertainty estimation, calibration under shift, source-action policies, and evaluation beyond accuracy. Long-term work should support deployment in broader modalities, multi-turn and agentic settings, and human-facing applications, with appropriate communication and governance mechanisms.

### *9.1. Source-aware uncertainty decomposition*

The field needs methods that separate perception, grounding, reasoning, answerability, and shift instead of collapsing them into one confidence value. This is not only a diagnostic preference. It changes the action. Visual ambiguity may call for a better input, grounding failure may call for evidence checking, reasoning instability may call for self-verification, and answerability failure may call for IDK or clarification. Future benchmarks should therefore label uncertainty sources when possible and test whether methods identify the right source, not only whether they predict correctness.

### *9.2. Action-aware benchmarks*

Benchmarks should evaluate whether uncertainty leads to better behavior. This requires tasks where the desired output may be an answer, an abstention, a clarification, a retrieval step, a prediction set, a self-check, or an escalation. Existing work on selective



用户不知道它是否反映了视觉基础、可回答性或语言模型的流利性。

对于面向人类的系统，不确定性沟通应具体且以行动为导向。模型应区分输入质量较差、证据缺失、模态冲突、推理不稳定以及高部署风险。它还应向用户提供有用的下一步操作：上传更清晰的图像、指定目标对象、提供缺失的上下文、允许检索、咨询专家，或将答案视为暂定。这种框架将不确定性估计与信任校准和问责联系起来。

这些部署压力暴露了该领域的主要差距：方法必须变得更加源感知、更加动作感知、在变化下更具稳健性，并且更容易在不同模型和模态之间复现。

## 9. 开放问题与研究议程

研究议程直接遵循源-信号-校准-行动链。近期优先事项是使不确定性评估更易于比较。中期优先事项是使不确定性估计能够意识到来源，并在现实访问限制下保持稳健。长期优先事项是将不确定性整合到交互式、使用工具的以及面向人类的多模态系统中。图6将这些优先事项总结为时间范围路线图。

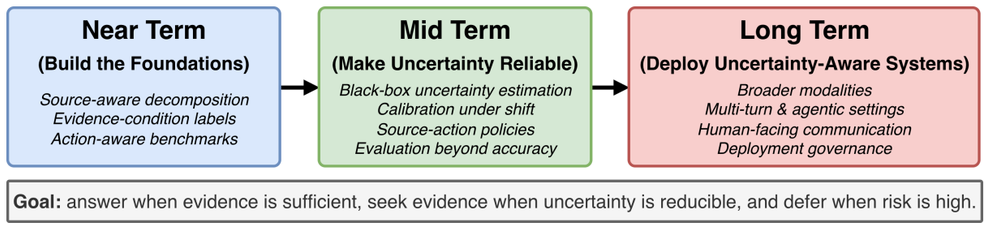



图6：面向不确定性感知大型多模态模型（MLLM）的研究路线图。该路线图按时间范围组织未来工作。近期工作应建立源感知不确定性分析的基础，包括证据-条件标签和动作感知基准。中期工作应通过黑箱不确定性估计、在变化下的校准、源-动作策略以及超越准确率的评估，使不确定性估计在现实访问和部署条件下可靠。长期工作应支持在更广泛的模态、多轮和智能体环境以及面向人类的应用中的部署，并配备适当的沟通和治理机制。

### *9.1.* 源感知的不确定性分解

该领域需要能够将感知、基础、推理、可回答性和转移分离的方法，而不是将它们合并为一个置信值。这不仅仅是一个诊断上的偏好。它会改变行动。视觉模糊可能需要更好的输入，基础失败可能需要证据核查，推理不稳定可能需要自我验证，而可回答性失败可能需要回答“我不知道”或请求澄清。因此，未来的基准测试应尽可能标注不确定性来源，并测试方法是否能够识别正确的来源，而不仅仅是预测正确性。

### *9.2.* 行为感知基准

基准测试应评估不确定性是否会导致更好的行为。这需要任务，其期望输出可能是一个答案、一次回避、一种澄清、一个检索步骤、一组预测、自我检查或升级。现有关于选择性的工作

answering, evidence collection, conformal abstention, and risk-controlled response selection points in this direction [71, 17, 16, 18]. The next step is to make action costs explicit so that methods can be compared by downstream utility, not only by correlation with correctness.

### *9.3. Robust calibration under shift and black-box access*

Many deployed MLLMs expose only text outputs. Researchers may not have logits, hidden states, image features, gradients, or attention maps. This makes black-box uncertainty central: repeated sampling, semantic disagreement, visual perturbation, verbalized confidence, external verifiers, and judge scores are often among the feasible signals. At the same time, calibration can fail under changes in domain, image quality, modality, language, prompt format, or user population [7, 10, 47, 64]. Future work should report model-access assumptions and calibration distributions explicitly, especially when claims are intended for deployment.

### *9.4. Broader modality and interaction coverage*

The evidence base remains most concentrated in image-text settings. Audio, video, multi-image reasoning, multi-turn dialogue, documents, charts, embodied systems, and tool-using agents require more systematic uncertainty evaluation. Each setting introduces uncertainty sources that are hard to reduce to static image QA: temporal absence in video, acoustic noise in audio, layout and numerical extraction in charts and documents, partial observability in embodied systems, and compounding decisions in agents. Broader modality coverage should therefore include source labels and action policies, not only new accuracy benchmarks.

### *9.5. Reproducibility and human-centered communication*

The field needs reporting standards that make results cumulative. At minimum, papers should report the model and version, modality, benchmark, prompt, model-access level, calibration data, uncertainty signal, confidence target, action policy, thresholds, cost assumptions, metrics, and failure cases. When possible, authors should release prompts, calibration splits, evaluator instructions, and evidence matrices. Reproducibility is also tied to user communication: future studies should test how people interpret confidence, hedging, abstention, clarification, and escalation, and whether those signals improve human decision quality rather than merely shifting responsibility.

The agenda therefore points toward decision-centered uncertainty. Estimation and calibration remain essential, but the field should increasingly ask whether uncertainty improves behavior: Does the model abstain when evidence is insufficient? Does it ask the right clarification? Does it retrieve useful evidence? Does it escalate high-risk cases? Does it communicate uncertainty in a way that helps humans act? These questions define the transition from uncertainty measurement to uncertainty-aware multimodal decision making.

## 10. Limitations of This Survey

This survey has several limitations. First, the evidence base is time-bounded. Work on MLLM uncertainty, abstention, calibration, and risk control is developing quickly, and several relevant studies remain available as preprints or under active revision. Reported findings may change as models, benchmarks, venues, and paper versions are updated. For



回答、证据收集、符合性回避以及风险控制的响应选择都指向这个方向 [71, 17, 16, 18]。下一步是明确行动成本，以便方法可以通过下游效用进行比较，而不仅仅是通过与正确性的相关性。

### *9.3.* 在分布变化和黑箱访问下的稳健校准

许多已部署的多模态大语言模型（MLLMs）仅提供文本输出。研究人员可能无法获取对数几率、隐藏状态、图像特征、梯度或注意力图。这使得黑箱不确定性成为核心：重复采样、语义分歧、视觉扰动、口头化置信度、外部验证器和评分往往是可行的信号。同时，当领域、图像质量、模态、语言、提示格式或用户群体发生变化时，校准可能会失败 [7, 10, 47, 64]。未来的工作应明确报告模型访问假设和校准分布，尤其是在声称用于部署的情况下。

### *9.4.* 更广的模态和交互覆盖

现有的证据基础仍然主要集中在图像-文本环境中。音频、视频、多图像推理、多轮对话、文档、图表、具身系统和使用工具的智能体需要更系统的不确定性评估。每种环境都会引入难以简化为静态图像问答的不确定性来源：视频中的时间缺失，音频中的声学噪声，图表和文档中的布局和数值提取，具身系统中的部分可观测性，以及智能体中的累积决策。因此，更广泛的模态覆盖应包括来源标签和行动策略，而不仅仅是新的准确性基准。

### *9.5.* 可重复性与以人为本的沟通

该领域需要能够使结果累积的报告标准。至少，论文应报告模型及其版本、模态、基准、提示、模型访问级别、校准数据、不确定性信号、置信目标、行动策略、阈值、成本假设、指标以及失败案例。在可能的情况下，作者应发布提示、校准分割、评估者说明和证据矩阵。可重复性也与用户通信相关：未来的研究应测试人们如何解读置信、保留意见、回避、澄清和升级，以及这些信号是否提高了人类决策质量，而不仅仅是转移责任。

因此，议程指向以决策为中心的不确定性。估计和校准仍然至关重要，但该领域应越来越多地问：不确定性是否改善了行为？当证据不足时，模型是否会选择回避？它是否提出了正确的澄清问题？它是否检索到有用的证据？它是否对高风险案例进行升级处理？它是否以帮助人类行动的方式传达不确定性？这些问题定义了从不确定性测量到不确定性感知的多模态决策的过渡。

## 10. 本次调查的局限性

本次调查存在若干局限性。首先，证据基础具有时间限制。关于多语言大模型的不确定性、回避、校准和风险控制的研究正在迅速发展，而且一些相关研究仍以预印本形式存在或正在积极修订中。随着模型、基准、会议和论文版本的更新，所报告的发现可能会发生变化。对于

this reason, we avoid claims that depend on a fixed leaderboard position or on a single reported number unless the comparison is made within the original study.

Second, coverage is uneven across modalities. Image-text settings are better represented than audio, video, multi-image dialogue, embodied systems, omni-modal inputs, and long-horizon agentic workflows. This imbalance reflects the current evidence base and should not be read as a judgment that those settings are less important. In fact, the weaker coverage of these modalities is one reason the research agenda emphasizes broader modality and interaction coverage.

Third, the survey is a structured synthesis rather than a quantitative meta-analysis. The papers differ in model access, prompt design, calibration data, score functions, evaluation targets, thresholds, and action policies. These differences make direct numerical comparison difficult. We therefore use tables and figures to organize concepts, assumptions, and decision logic, while leaving raw corpus counts and search-protocol details to the appendix or companion supplement.

These limitations do not weaken the need for the source–signal–calibration–action framework. They clarify its role. The framework is not a substitute for domain-specific validation; it is a way to make uncertainty claims easier to locate, compare, and audit.

## 11. Conclusion

MLLM uncertainty is broader than confidence in a generated sentence. A response can be fluent while the evidence is blurred, absent, visually misread, weakly grounded, contradicted by another modality, unstable under reasoning, shifted from the calibration domain, or too risky for autonomous use. Treating these cases as one generic confidence problem hides the reason the system may fail and obscures what it should do next.

This survey organized the field around a source–signal–calibration–action chain. Sources explain where unreliability enters the multimodal pipeline. Signals make those sources observable through probabilities, representations, samples, perturbations, grounding checks, verbalized confidence, verifier scores, judge models, or conformal scores. Calibration and risk control give these signals decision meaning. Actions then determine whether the system answers, hedges, abstains, asks for clarification, requests better input, retrieves evidence, self-checks, routes, or escalates.

The main conclusion is that uncertainty-aware MLLMs should be evaluated by behavior under uncertainty, not by confidence quality alone. Progress will require source-aware decomposition, action-aware benchmarks, calibration that survives shift and black-box access, broader coverage of audio, video, documents, charts, embodied agents, and interactive settings, and communication strategies that help users understand both the uncertainty and the next step. The next generation of uncertainty-aware MLLMs should not merely know when they are uncertain; they should know what to do because of that uncertainty.

## Appendix A. Evidence Mapping and Traceability

The appendix retains procedural material that supports auditability but would be too heavy for the main text: search sources, query families, screening counts, evidence-tier definitions, extraction fields, and corpus-refresh details. This material documents how the evidence matrix was assembled without making the search procedure the center of the survey. The evidence mapping distinguishes the 67 focused evidence rows from background survey citations and from non-counted bridge material retained for traceability.



基于这个原因，我们避免依赖固定排行榜位置或单一报告数字的声明，除非比较是在原始研究中进行的。

其次，各种模态的覆盖是不均衡的。图文设置的代表性要优于音频、视频、多图像对话、具身系统、全模态输入和长期代理工作流程。这种不平衡反映了当前的证据基础，不应被解读为这些设置不重要。实际上，这些模态覆盖较弱正是研究议程强调更广泛模态和交互覆盖的一个原因。

第三，本次调查是一个结构化的综合，而非定量的元分析。各篇论文在模型访问、提示设计、校准数据、评分函数、评估目标、阈值以及行动策略上存在差异。这些差异使直接的数值比较变得困难。因此，我们使用表格和图表来组织概念、假设和决策逻辑，同时将原始语料计数和搜索协议的详细信息留到附录或附带补充材料中。

这些限制并不削弱对源–信号–校准–行动框架的需求。它们澄清了该框架的作用。该框架不是特定领域验证的替代品；它是一种使不确定性声明更容易定位、比较和审计的方法。

## 11. 结论

MLLM的不确定性比生成句子的置信度更广泛。一个回答可以是流利的，但其证据可能模糊、缺失、视觉上被误读、基础薄弱、被另一种模态所矛盾、在推理下不稳定、偏离校准领域，或对于自主使用来说风险过大。将这些情况视为一个通用的置信度问题会掩盖系统可能失败的原因，并模糊它接下来应该做什么。

这项调查将该领域组织在一个源–信号–校准–行动链条中。源解释了不可靠性如何进入多模态流程。信号通过概率、表示、样本、扰动、验证检查、口头表达的置信度、验证者评分、评审模型或一致性评分使这些源可观察。校准和风险控制赋予这些信号决策意义。然后，行动决定系统是回答、保留、回避、请求澄清、请求更好的输入、检索证据、自检、路由还是升级处理。

主要结论是，不确定性感知的多模态大模型（MLLM）应通过在不确定性下的表现来评估，而不仅仅是通过置信度质量来评估。要取得进展，需要源感知的分解、动作感知的基准测试、能够在分布变化下仍然有效的校准和黑箱访问、对音频、视频、文档、图表、具身代理和交互环境更广泛的覆盖，以及帮助用户理解不确定性和下一步操作的沟通策略。新一代不确定性感知的多模态大模型不仅应知道何时存在不确定性；它们还应知道应因这种不确定性而采取的措施。

## 附录A. 证据映射与可追溯性

附录保留了支持可审计性的程序性材料，但这些材料放在正文中会过于冗重：搜索来源、查询类别、筛选计数、证据层级定义、提取字段以及语料库更新细节。这些材料记录了证据矩阵的组装方式，而不会将搜索程序作为综述的核心。证据映射区分了 67 行重点证据、背景调查引用以及为可追溯性保留的未计数桥接材料。

Main-table citations are representative row-level anchors, not a replacement for the full evidence matrix. A final source-verification pass also flags venue status and withdrawn or actively revised preprints so that the main text does not rely on unsupported publication metadata.

## Appendix B. Main-Text and Supplement Boundary

The main text is limited to five synthesis tables and six conceptual figures. Additional traceability artifacts are retained here or in the companion supplement: corpus composition, evidence tiers, targeted evidence-audit summaries, search-log fields, representative evidence excerpts, evidence-field schema, metric-family intersection counts, source-action heatmap counts, application-domain risk diagnostics, and count-heavy research-roadmap diagnostics. These materials support reproducibility without competing with the main conceptual story.

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主表引用是具有代表性的行级锚点，而不是完整证据矩阵的替代。最终的来源验证过程也会标记出版地点状态以及已撤回或正在修改的预印本，以确保正文不依赖未经支持的出版元数据。

## 附录 B. 正文与补充部分的界限

正文限于五张综合表和六个概念图。其他可追溯性材料保留在此处或伴随的补充材料中：语料库组成、证据等级、目标证据审计摘要、搜索日志字段、代表性证据摘录、证据字段模式、指标系列交集计数、来源-操作热图计数、应用领域风险诊断以及计数密集型研究路线图诊断。这些材料支持可重复性，同时不与主要概念故事竞争。

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